

Week 2 – Dataset Exploration and First Wrangling

Selecting, Cleaning, and Understanding Your Data

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PUBLISHED

November 18, 2025

1 Overview

Introduce the purpose of Week 2 and what this document covers.

- Key aims of this week
 - Define objectives of the project and choose appropriate data from source.
 - State the hypothesis from the objectives and justify the obtained data.
 - Perform data wrangling (Duplicate identification, Missing data, and data transformation)
 - Perform basic EDA (structure, summaries, missingness, simple plots) and record issues/questions it reveals
 - Discuss the reflection from each section with the team.
- Expected outcomes
 - Project objective finalized and extract raw data from source.
 - Go through different objective and literature and fixed the hypothesis.
 - Raw data has been transformed into data for preliminary analysis.
 - Using basic EDA, try to establish data consistency, and patterns for next level deeper analysis
 - Write down our learning and establish best way practices for the project
- General encouragement statement

The project gives us the opportunity to learn and implement our learning and skills as a data engineer and understand the workflow the process in its entirety.

2 Dataset Selection and Rationale

2.1 1.1 Dataset choice

Transport industry has been of the largest contributors greenhouse gas(GHG) emission. The primary goal in reducing emissions is vital to reach the EU's 2050 climate neutrality objective. Electric vehicles(EV), which include battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), are penetrating the EU market to adhere to this objective. The main objective of our study is to determine various socio-economic determinants for EV adoption in EU and to quantify the relative importance of these variables.

The data set provides comprehensive information on global electric vehicle (EV) trends, focusing on vehicle stock, sales, charging infrastructure, and powertrain types (BEV, PHEV, FCEV) from 2010 to 2024. It includes detailed country-level data across multiple indicators such as EV adoption, policy incentives, energy use, and charging point deployment. The data set enables our team to understand various aspects of our learning from different modules and have an analytical report of the comparative analysis of EV growth, particularly across European region.

2.1.1 (Insert code cells – load dataset)

▼ Show code

```
# Step 0: Load Required Libraries
```

```
options(dplyr.summarise.inform = TRUE)
knitr::opts_chunk$set(fig.width = 12,
  fig.height = 8,
  include = TRUE,
  echo = TRUE)
```

```
library(tidyverse)
library(readxl)
library(dplyr)
```

```
# Step 1: Load the Dataset
```

```
# Reads the first sheet ("Page 1" in Excel)
```

```
ev_data <- read_excel("data/raw/EVDataExplorer2025.xlsx", sheet = 1)
#ev_data <- read_excel("~/Desktop/R_data/MA7441/Data/raw/EVDataExplorer2025.xlsx", she
head(ev_data, 10)
```

region_country	category	parameter	mode	powertrain	y
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
World	Projection-STEPS	EV stock	2 and 3 wheelers	BEV	2025
World	Projection-STEPS	EV stock	Cars	BEV	2025
China	Projection-STEPS	EV stock	2 and 3 wheelers	BEV	2025
China	Projection-STEPS	EV stock	Cars	BEV	2025
World	Projection-STEPS	EV stock	Cars	PHEV	2025
World	Historical	EV stock	2 and 3 wheelers	BEV	2025
World	Projection-STEPS	EV stock	2 and 3 wheelers	BEV	2025
Asia Pacific	Historical	EV stock	2 and 3 wheelers	BEV	2025
World	Historical	EV stock	2 and 3 wheelers	BEV	2025
World	Projection-STEPS	EV stock	2 and 3 wheelers	BEV	2025

1-10 of 10 rows | 1-6 of 9 columns

2.2 1.2 Justification

The key objective of this study is try to quantify the influence of demographic, economic, and infrastructural variables on EV adoption rates. The main hypothesis of our study which was established through extensive literature surveys where EV adoption rate determinants was investigated. Smith et al. (2022) found that income and urban density are significant predictors, while charging infrastructure availability strongly influences consumer decisions.

H1 Higher median income positively correlates with EV adoption. H2 Greater density of public charging stations leads to higher adoption. H3 Regions with higher fuel prices have faster EV uptake.

As an preliminary step, we focus on H2 hypothesis. The obtained data includes country-level which includes EU27 data on EV sales(BEV, PHEV, FCEV) and charging point deployment. This helps us make a preliminary analysis of the H2 hypothesis. The study will extend to other variables for our final report. We primarily focus on last 10 years of data(2015-2024), before which, the overall data seems to be sparse for many countries.

3 2 First Glimpse of the Data

3.1 2.1 Initial inspection

Discuss the structure, dimensions, and variable types

- The data frame contains 9 columns
- The data frame has 16,436 row (observations)

3.1.1 (Insert code cells – glimpse, summary)

▼ Show code

```
# Step 2: Initial inspection
glimpse(ev_data)
```

Rows: 16,436

Columns: 9

```
$ region_country    <chr> "World", "World", "China", "China", "World", "World"...
$ category          <chr> "Projection-STEPS", "Projection-STEPS", "Projection-...
$ parameter         <chr> "EV stock", "EV stock", "EV stock", "EV stock", "EV ...
$ mode              <chr> "2 and 3 wheelers", "Cars", "2 and 3 wheelers", "Car...
$ powertrain        <chr> "BEV", "BEV", "BEV", "BEV", "PHEV", "BEV", "BEV", "B...
$ year              <dbl> 2030, 2030, 2030, 2030, 2030, 2024, 2024, 2024, 2023...
$ unit              <chr> "Vehicles", "Vehicles", "Vehicles", "Vehicles", "Veh...
$ value             <dbl> 1.7e+08, 1.5e+08, 9.1e+07, 8.2e+07, 8.2e+07, 7.9e+07...
$ `Aggregate group` <chr> "_World", "_World", "Other", "Other", "_World", "_Wo..."
```

▼ Show code

```
summary(ev_data)
```

region_country	category	parameter	mode
Length:16436	Length:16436	Length:16436	Length:16436
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

powertrain	year	unit	value
Length:16436	Min. :2010	Length:16436	Min. : 0
Class :character	1st Qu.:2016	Class :character	1st Qu.: 1
Mode :character	Median :2019	Mode :character	Median : 49
	Mean :2019		Mean : 267827
	3rd Qu.:2022		3rd Qu.: 2500

Max. : 2030

Max. : 170000000

Aggregate group
 Length: 16436
 Class : character
 Mode : character

3.2 2.2 Observations

Summarise key insights about shape, columns, and data types.

- The data frame contains 9 columns, which includes 'region_country', 'category', 'parameter', 'mode', 'powertrain', 'year', 'unit', 'value', 'Aggregate group'. The main parameters we are interested are the region_country, year and unit.
- The data frame has 16,436 row, which has lot of information. but we focus the EU countries, EV stock data, Cars for the EV adoption and EV charging point data.
- So, The shape of a data frame is (16436,9)
- The variable are either Character or double. There are 7 character tpye variables and 2 doubles type variable.
- The statistical information from value is not inferenceable as it contains information from various range of categories and has to be deconstructed into various groups and then analysed.
- The statistical information from year tells us that, historical data is available from 2010 to 2024 and projected data is available till 2030

4 3 Initial Data Wrangling

Document cleaning, transformations, and reasoning.

The data set is fairly decent with no missing information rro duplication, as the source is a. well recorded government agency. Steps are,

- checks for missing data
- check for duplication
- Format check in year, countries and value

4.1 3.1 Cleaning

Describe how you handled missing values, duplicates, formats.

(Insert code cells – cleaning steps)

▼ Show code

```
# 3.1.1 : Count missing values in each column
colSums(is.na(ev_data))
```

region_country	category	parameter	mode	powertrain
0	0	0	0	0
year	unit	value	Aggregate	group
0	0	0	0	0

▼ Show code

```
# 3.1.2 : Remove rows with any missing values
ev_data_clean <- na.omit(ev_data)

# 3.1.3 : formating
ev_data$region_country <- str_trim(ev_data$region_country)
```

4.2 3.2 Transformation, integration, validation

Conceptual or implemented; note decisions or possible future steps.

The data has to be cleaned and transformed for the next step of analysis. We need to extract just the EV sales, Electric charging points, FCEV, PHEV data from 2015 to 2024 for only the European countries. The following steps were followed,

- Identify the list of European countries and store it in a vector called europe_countries.
- Using filtering process filter out the European countries, and str_detect function filter put only historical, BEV, EV stock, Cars data from year 2015 to 2024.
- Selecting only the required columns
- Cleaning the names of the data from any white spaces
- Grouping and summary to see the data
- using Pivot wide function to create a table with different years and countries.

This process has been repeated from different parameters that we plan to study like Electric charging points, FCEV, PHEV.

Critical Decision parameters: - European countries which has only the significant data - Pivot wide to make it 3d table rather than 2d for analysis - Historical data alone, No need to have projections. - BEV cars only for now (as it is more significant in decision implementations)

Possible future step: This has to be done for other future data set such as population, income levels, fuel prices.

(Insert code cells – optional transformation)

▼ Show code

```
# Step 3.2.1: Define List of European Countries
europe_countries <- c(
  "Austria", "Belgium", "Bulgaria", "Croatia", "Cyprus", "Czech Republic",
  "Denmark", "Estonia", "Finland", "France", "Germany", "Greece", "Hungary",
  "Ireland", "Italy", "Latvia", "Lithuania", "Luxembourg", "Malta",
  "Netherlands", "Poland", "Portugal", "Romania", "Slovakia", "Slovenia",
  "Spain", "Sweden", "Norway", "Iceland", "Switzerland", "United Kingdom"
)

# Step 3.2.2: EV Sales for year 2015 – 2024
```

```
# Step 3.2.2.1: Filter for Europe, BEV, and Years 2015–2024
ev_europe_bev <- ev_data %>%
  filter (
    region_country %in% europe_countries,      # Only European countries
    str_detect(powertrain, regex("BEV", ignore_case = TRUE)), # BEV only
    year >= 2015 & year <= 2024,               # Year range
    str_detect(parameter, regex("EV stock", ignore_case = TRUE)), # EV stock data only
    str_detect(category, regex("Historical", ignore_case = TRUE)), # His. data
    str_detect(mode, regex("Cars", ignore_case = TRUE)) # Passenger cars only
  )

# Step 3.2.2.2: Select Useful Columns
ev_europe_bev <- ev_europe_bev %>%
  select(region_country, year, value, unit)

# Step 3.2.2.3: Clean Country Names (trim whitespace)
ev_europe_bev$region_country <- str_trim(ev_europe_bev$region_country)

# Step 3.2.2.4: filter and summary
filtered_ev_data <- ev_europe_bev |>
  group_by(region_country, year) |>
  select(region_country, year, value)
summary(filtered_ev_data)
```

region_country	year	value
Length:170	Min. :2015	Min. : 97
Class :character	1st Qu.:2017	1st Qu.: 8725
Mode :character	Median :2020	Median : 35500
	Mean :2020	Mean : 149446
	3rd Qu.:2022	3rd Qu.: 160000
	Max. :2024	Max. :1900000

▼ Show code

```
# Step 3.2.2.5: Pivot: Countries as rows, Years as columns
ev_wide <- filtered_ev_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)
```

▼ Show code

```
# Step 3.2.3: EV Charging for year 2015 – 2024 for different countries
# Step 3.2.3.1: Filter for Europe, , and Years 2015–2024
ec_europe_bev <- ev_data %>%
  filter (
    region_country %in% europe_countries,      # Only European countries
    #str_detect(powertrain, regex("EV", ignore_case = TRUE)), # BEV only
    year >= 2015 & year <= 2024,               # Year range
    str_detect(parameter, regex("EV charging points", ignore_case = TRUE)), # EV stock
    str_detect(category, regex("Historical", ignore_case = TRUE)), # His. data
  )
```

```

    str_detect(mode, regex("EV", ignore_case = TRUE))# Passenger cars only
  )

# Step 3.2.3.2: Select Useful Columns
ec_europe_bevev <- ec_europe_bevev %>%
  select(region_country, year, value, unit)

# Step 3.2.3.3: Clean Country Names (trim whitespace)
ec_europe_bevev$region_country <- str_trim(ec_europe_bevev$region_country)

# Step 3.2.3.4: filter and summary
filtered_ec_data <- ec_europe_bevev |>
  group_by(region_country, year) |>
  select(region_country, year, value)
summary(filtered_ec_data)

```

region_country	year	value
Length:325	Min. :2015	Min. : 1
Class :character	1st Qu.:2017	1st Qu.: 580
Mode :character	Median :2020	Median : 2800
	Mean :2020	Mean : 10503
	3rd Qu.:2022	3rd Qu.: 9800
	Max. :2024	Max. :180000

▼ Show code

```

# Step 3.2.3.5: Pivot: Countries as rows, Years as columns
ec_wide <- filtered_ec_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)

```

▼ Show code

```

# Step 3.2.4: FEV for year 2015 – 2024 for different countries
ev_europe_fev <- ev_data %>%
  filter (
    region_country %in% europe_countries,          # Only European countries
    str_detect(powertrain, regex("FCEV", ignore_case = TRUE)), # FCEV only
    year >= 2015 & year <= 2024,                    # Year range
    str_detect(parameter, regex("EV stock", ignore_case = TRUE)),# EV stock data only
    str_detect(category, regex("Historical", ignore_case = TRUE)),# His. data
    str_detect(mode, regex("Cars", ignore_case = TRUE))# Passenger cars only
  )

# Step 3.2.4.1: Select Useful Columns
ev_europe_fev <- ev_europe_fev %>%
  select(region_country, year, value, unit)

# Step 3.2.4.2: Clean Country Names (trim whitespace)
ev_europe_fev$region_country <- str_trim(ev_europe_fev$region_country)

```

```
# Step 3.2.4.3: filter and summary
filtered_fev_data <- ev_europe_fev |>
  group_by(region_country, year) |>
  select(region_country, year, value)
summary(filtered_fev_data)
```

```
region_country      year      value
Length:142         Min.   :2015   Min.    :  1.00
Class :character    1st Qu.:2018   1st Qu.: 23.25
Mode  :character    Median :2020   Median : 52.50
                  Mean   :2020   Mean   :189.17
                  3rd Qu.:2022   3rd Qu.:210.00
                  Max.   :2024   Max.   :2300.00
```

▼ Show code

```
# Step 3.2.4.4: Pivot: Countries as rows, Years as columns
fev_wide <- filtered_fev_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)

#Step 3.2.4.5 :Count missing values in each column
colSums(is.na(fev_wide))
```

```
region_country      2015      2016      2017      2018
0                5        5        3        2
2019             2020      2021      2022      2023
1                1        0        0        0
2024
1
```

▼ Show code

```
# Step 3.2.5: PHEV for year 2015 – 2024 for different countries
ev_europe_phev <- ev_data %>%
  filter (
    region_country %in% europe_countries, # Only European countries
    str_detect(powertrain, regex("PHEV", ignore_case = TRUE)), # FCEV only
    year >= 2015 & year <= 2024, # Year range
    str_detect(parameter, regex("EV stock", ignore_case = TRUE)), # EV stock data only
    str_detect(category, regex("Historical", ignore_case = TRUE)), # His. data
    str_detect(mode, regex("Cars", ignore_case = TRUE)) # Passenger cars only
  )

# Step 3.2.5.1: Select Useful Columns
ev_europe_phev <- ev_europe_phev %>%
  select(region_country, year, value, unit)

# Step 3.2.5.2: Clean Country Names (trim whitespace)
ev_europe_phev$region_country <- str_trim(ev_europe_phev$region_country)
```



```
# Step 3.2.5.3: filter and summary
filtered_phev_data <- ev_europe_phev |>
  group_by(region_country, year) |>
  select(region_country, year, value)
summary(filtered_phev_data)
```

```
region_country      year      value
Length:166         Min.   :2015   Min.    :    22
Class :character    1st Qu.:2017   1st Qu.:   9650
Mode  :character    Median :2020   Median :  45000
                        Mean  :2020   Mean   : 112499
                        3rd Qu.:2022   3rd Qu.: 120000
                        Max.   :2024   Max.   :1200000
```

▼ Show code

```
# Step 3.2.5.4: Pivot: Countries as rows, Years as columns
phev_wide <- filtered_phev_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)

#Step 3.2.5.5 :Count missing values in each column
colSums(is.na(phev_wide))
```

```
region_country      2015      2016      2017      2018
0                  1          1          1          1
2019              2020      2021      2022      2023
0                  0          0          0          0
2024
0
```

4.3 3.3 Storing cleaned data

Describe where you saved the processed file and why.

The all processed data has been stored in processed data set.

(Insert code cells – write processed file)

▼ Show code

```
# Step 7: Save the Filtered Dataset
path <- "data/processed/"
write_csv(filtered_ev_data, file.path(path, "EV_Europe_BEV_Stock_2015_2024.csv"))
write_csv(filtered_ec_data, file.path(path, "EC_Europe_BEV_Stock_2015_2024.csv"))
write_csv(filtered_fev_data, file.path(path, "EV_Europe_FEV_Stock_2015_2024.csv"))
write_csv(filtered_phev_data, file.path(path, "EV_Europe_PHEV_Stock_2015_2024.csv"))
```

5 4 Exploratory Data Analysis (EDA)

5.1 4.1 Variable identification

List variable types and their roles.

The variable are either Character or double. There are 1 character type variables and 2 doubles type variable. and other data frames have 1 character type variables and 11 doubles type variable. - Value is the most important variable as it lists the no. of each parameter and would be used for our analysis. - Data in value has categorized into for different year in other data from to form 3d vector rather than a 2d form for better visaulizaiton. - region_country can analysed categorically to check if all year has the data.

(Insert code cells – variable listing)

▼ Show code

```
# identify variable types

glimpse(filtered_ev_data)
```

Rows: 170

Columns: 3

Groups: region_country, year [170]

```
$ region_country <chr> "Germany", "Germany", "United Kingdom", "France", "Germ...
$ year           <dbl> 2024, 2023, 2024, 2024, 2022, 2023, 2023, 2024, 2023, 2...
$ value         <dbl> 1900000, 1500000, 1400000, 1100000, 1000000, 980000, 98...
```

▼ Show code

```
glimpse(ev_wide)
```

Rows: 17

Columns: 11

```
$ region_country <chr> "Austria", "Belgium", "Denmark", "Finland", "France", "...
$ `2015`        <dbl> 5000, 3300, 7900, 610, 45000, 26000, 97, 570, 4400, 940...
$ `2016`        <dbl> 9100, 5200, 8700, 840, 67000, 34000, 140, 960, 5800, 13...
$ `2017`        <dbl> 15000, 7500, 8800, 1400, 93000, 54000, 190, 1700, 7800,...
$ `2018`        <dbl> 21000, 11000, 10000, 2400, 120000, 83000, 280, 2400, 13...
$ `2019`        <dbl> 30000, 19000, 16000, 4700, 170000, 140000, 470, 3500, 2...
$ `2020`        <dbl> 45000, 32000, 32000, 9700, 280000, 310000, 1100, 5400, ...
$ `2021`        <dbl> 77000, 52000, 67000, 23000, 440000, 620000, 3300, 8500,...
$ `2022`        <dbl> 110000, 90000, 110000, 45000, 620000, 1000000, 7400, 14...
$ `2023`        <dbl> 160000, 190000, 200000, 84000, 980000, 1500000, 13000, ...
$ `2024`        <dbl> 200000, 320000, 340000, 110000, 1100000, 1900000, 18000...
```

▼ Show code

```
glimpse(ec_wide)
```

Rows: 17

Columns: 11

```
$ region_country <chr> "Austria", "Belgium", "Denmark", "Finland", "France", "...
$ `2019`              <dbl> 3722, 6590, 4960, 3490, 30700, 29200, 64, 260, 9400, 47...
$ `2020`              <dbl> 8600, 8740, 870, 4330, 34300, 43300, 228, 500, 13500, 6...
$ `2021`              <dbl> 10400, 12720, 890, 6590, 54900, 59500, 1228, 680, 22200...
$ `2022`              <dbl> 13000, 24200, 11400, 9400, 91000, 86000, 3080, 1030, 38...
$ `2023`              <dbl> 18600, 44000, 17000, 12200, 119000, 110000, 5160, 1810,...
$ `2024`              <dbl> 27300, 74600, 33400, 16400, 152000, 157000, 6000, 3040,...
$ `2015`              <dbl> NA, 1400, 2400, 840, 10480, 5070, 16, 31, 1770, 18220, ...
$ `2016`              <dbl> NA, 1640, 4540, 850, 20000, 7500, 15, 51, 2470, 32400, ...
$ `2017`              <dbl> NA, 1800, 4590, 850, 21000, 11600, 22, 68, 2860, 33410,...
$ `2018`              <dbl> NA, 3020, 4700, 2220, 25300, 19900, 32, 95, 3600, 36820...
```

▼ Show code

```
glimpse(fev_wide)
```

Rows: 16

Columns: 11

```
$ region_country <chr> "Austria", "Belgium", "Denmark", "Finland", "France", "...
$ `2015`          <dbl> 6, 8, 37, NA, 10, 86, 4, 11, 26, 13, NA, NA, NA, 4, 10,...
$ `2016`          <dbl> 13, 20, 68, NA, 34, 49, 3, 11, 33, 40, NA, NA, NA, 7, 3...
$ `2017`          <dbl> 19, 31, 83, NA, 29, 310, 2, 11, 39, 70, 14, NA, NA, 21,...
$ `2018`          <dbl> 24, 44, 88, NA, 90, 470, 17, 12, 50, 120, 28, NA, 1, 21...
$ `2019`          <dbl> 41, 41, 100, 1, 380, 640, 31, 28, 210, 180, 34, NA, 2, ...
$ `2020`          <dbl> 45, 64, 150, 1, 380, 1000, 30, 35, 380, 200, 28, NA, 9,...
$ `2021`          <dbl> 55, 85, 230, 1, 400, 1500, 31, 45, 540, 230, 91, 3, 19,...
$ `2022`          <dbl> 62, 80, 230, 2, 590, 2200, 31, 56, 640, 260, 130, 3, 23...
$ `2023`          <dbl> 62, 99, 230, 2, 960, 2100, 27, 59, 700, 290, 210, 3, 27...
$ `2024`          <dbl> 62, 100, 230, 1, 1600, 2300, NA, 60, 710, 230, 220, 3, ...
```

▼ Show code

```
glimpse(phev_wide)
```

Rows: 17

Columns: 11

```
$ region_country <chr> "Austria", "Belgium", "Denmark", "Finland", "France", "...
$ `2015`          <dbl> 1500, 3800, 570, 970, 9300, 19000, 22, 140, 1300, 79000...
$ `2016`          <dbl> 2300, 11000, 1100, 2400, 17000, 32000, 78, 600, 2700, 9...
$ `2017`          <dbl> 3900, 23000, 1800, 5700, 54000, 50000, 220, 2200, 5600,...
$ `2018`          <dbl> 5700, 33000, 5200, 13000, 68000, 82000, 440, 5600, 1000...
$ `2019`          <dbl> 8000, 41000, 9900, 25000, 85000, 100000, 730, 7500, 170...
$ `2020`          <dbl> 15000, 73000, 30000, 46000, 160000, 300000, 2200, 9600,...
$ `2021`          <dbl> 29000, 120000, 78000, 77000, 300000, 630000, 7000, 1300...
$ `2022`          <dbl> 42000, 180000, 100000, 100000, 430000, 860000, 12000, 1...
$ `2023`          <dbl> 58000, 280000, 110000, 140000, 590000, 1000000, 21000, ...
$ `2024`          <dbl> 75000, 350000, 120000, 150000, 650000, 1200000, 27000, ...
```

5.2 4.2 Descriptive statistics

Discuss central tendencies, dispersion, outliers.

EV sales growth

Across 2015–2024, the mean and median values both rise sharply, showing rapid market expansion.

Mean: increases from ~12,000 (2015) to ~473,000 (2024), indicating average EV stocks per country grew nearly 40×.

Median: follows a similar trend, from ~4,800 to ~300,000, showing that even mid-range adopters experienced massive growth.

Mode: not applicable (all unique country values).

Range: widens from ~59,000 – 97 (2015) to ~1.9 million – 18,000 (2024).

Interquartile range (IQR): expands from a few thousand in early years to hundreds of thousands by 2024.

Standard deviation (not shown, but implied): rises sharply — variability among countries is large, reflecting different national EV policies and infrastructure levels.

Germany (1.9M in 2024), France (1.1M), and UK (1.4M) likely sit far above others.

Statistically, these would be upper outliers if using the 1.5×IQR rule — but contextually, they represent legitimate leadership in EV adoption, not data errors.

(Insert code cells – summarise numeric variables)

▼ Show code

```
# Step 4.2.1 : Summary
summary(filtered_ev_data)
```

```
region_country      year      value
Length:170         Min.   :2015   Min.    :    97
Class :character    1st Qu.:2017   1st Qu.:   8725
Mode  :character    Median :2020   Median :  35500
                        Mean   :2020   Mean    : 149446
                        3rd Qu.:2022   3rd Qu.: 160000
                        Max.    :2024   Max.    :1900000
```

▼ Show code

```
summary(ev_wide)
```

```
region_country      2015      2016      2017
Length:17          Min.    :   97   Min.    :  140   Min.    :  190
Class :character    1st Qu.: 2500   1st Qu.: 2800   1st Qu.: 4700
Mode  :character    Median : 4800   Median : 7500   Median :10000
                        Mean   :11987   Mean   :16949   Mean   :24541
                        3rd Qu.: 9400   3rd Qu.:13000   3rd Qu.:21000
                        Max.    :59000   Max.    :83000   Max.    :120000
      2018      2019      2020      2021
Min.    :   280   Min.    :   470   Min.    :  1100   Min.    :  3300
1st Qu.:  8700   1st Qu.: 16000   1st Qu.: 23000   1st Qu.: 37000
Median : 16000   Median : 26000   Median : 45000   Median : 77000
Mean   : 34734   Mean   : 55204   Mean   : 98094   Mean   :165341
```

3rd Qu.: 45000	3rd Qu.: 91000	3rd Qu.:180000	3rd Qu.:250000
Max. :160000	Max. :220000	Max. :340000	Max. :620000
2022	2023	2024	
Min. : 7400	Min. : 13000	Min. : 18000	
1st Qu.: 54000	1st Qu.: 91000	1st Qu.: 130000	
Median : 110000	Median : 190000	Median : 300000	
Mean : 244553	Mean : 370118	Mean : 472941	
3rd Qu.: 340000	3rd Qu.: 460000	3rd Qu.: 590000	
Max. :1000000	Max. :1500000	Max. :1900000	

▼ Show code

```
summary(ec_wide)
```

region_country	2019	2020	2021
Length:17	Min. : 64	Min. : 228	Min. : 680
Class :character	1st Qu.: 3490	1st Qu.: 2920	1st Qu.: 4200
Mode :character	Median : 6320	Median : 8700	Median :10400
	Mean :11013	Mean :14806	Mean :20424
	3rd Qu.:14000	3rd Qu.:17300	3rd Qu.:22200
	Max. :47300	Max. :65300	Max. :83800
2022	2023	2024	2015
Min. : 1030	Min. : 1810	Min. : 3040	Min. : 16
1st Qu.: 9400	1st Qu.: 12200	1st Qu.: 16400	1st Qu.: 1035
Median : 13000	Median : 25600	Median : 33400	Median : 1705
Mean : 31208	Mean : 42081	Mean : 56696	Mean : 3901
3rd Qu.: 38000	3rd Qu.: 48100	3rd Qu.: 74600	3rd Qu.: 5185
Max. :124500	Max. :145000	Max. :188000	Max. :18220
			NA's :1
2016	2017	2018	
Min. : 15	Min. : 22	Min. : 32	
1st Qu.: 1218	1st Qu.: 1450	1st Qu.: 2110	
Median : 3065	Median : 3835	Median : 4770	
Mean : 6257	Mean : 7116	Mean : 8822	
3rd Qu.: 7500	3rd Qu.: 9815	3rd Qu.:11900	
Max. :32400	Max. :33410	Max. :36820	
NA's :1	NA's :1	NA's :1	

▼ Show code

```
summary(fev_wide)
```

region_country	2015	2016	2017
Length:16	Min. : 4.00	Min. : 3.00	Min. : 2.00
Class :character	1st Qu.: 7.00	1st Qu.:12.00	1st Qu.: 19.00
Mode :character	Median :10.00	Median :33.00	Median : 31.00
	Mean :19.55	Mean :28.55	Mean : 53.85
	3rd Qu.:19.50	3rd Qu.:38.00	3rd Qu.: 39.00
	Max. :86.00	Max. :68.00	Max. :310.00
	NA's :5	NA's :5	NA's :3
2018	2019	2020	2021
Min. : 1.00	Min. : 1.0	Min. : 1.0	Min. : 1.0
1st Qu.: 21.75	1st Qu.: 32.5	1st Qu.: 32.5	1st Qu.: 41.5

Median : 47.00	Median : 42.0	Median : 64.0	Median : 88.0
Mean : 78.14	Mean : 132.3	Mean : 178.9	Mean : 229.6
3rd Qu.: 83.50	3rd Qu.: 175.0	3rd Qu.: 215.0	3rd Qu.: 232.5
Max. : 470.00	Max. : 640.0	Max. : 1000.0	Max. : 1500.0
NA's : 2	NA's : 1	NA's : 1	

2022	2023	2024
Min. : 2.00	Min. : 2.00	Min. : 1.0
1st Qu.: 49.75	1st Qu.: 42.75	1st Qu.: 55.0
Median : 105.00	Median : 154.50	Median : 220.0
Mean : 298.25	Mean : 332.31	Mean : 407.3
3rd Qu.: 237.50	3rd Qu.: 267.50	3rd Qu.: 255.0
Max. : 2200.00	Max. : 2100.00	Max. : 2300.0
		NA's : 1

▼ Show code

```
summary(phev_wide)
```

region_country	2015	2016	2017
Length:17	Min. : 22	Min. : 78	Min. : 220
Class :character	1st Qu.: 870	1st Qu.: 2125	1st Qu.: 4050
Mode :character	Median : 2350	Median : 3900	Median : 7650
	Mean : 10464	Mean : 17661	Mean : 27614
	3rd Qu.: 9850	3rd Qu.: 22000	3rd Qu.: 51000
	Max. : 79000	Max. : 99000	Max. : 97000
	NA's : 1	NA's : 1	NA's : 1

2018	2019	2020	2021
Min. : 440	Min. : 730	Min. : 2200	Min. : 7000
1st Qu.: 7425	1st Qu.: 9900	1st Qu.: 26000	1st Qu.: 41000
Median : 13500	Median : 19000	Median : 44000	Median : 86000
Mean : 38809	Mean : 45937	Mean : 81594	Mean : 141353
3rd Qu.: 71500	3rd Qu.: 85000	3rd Qu.: 120000	3rd Qu.: 180000
Max. : 130000	Max. : 160000	Max. : 300000	Max. : 630000
NA's : 1			

2022	2023	2024
Min. : 12000	Min. : 21000	Min. : 27000
1st Qu.: 57000	1st Qu.: 84000	1st Qu.: 110000
Median : 130000	Median : 200000	Median : 210000
Mean : 190647	Mean : 249882	Mean : 300118
3rd Qu.: 200000	3rd Qu.: 270000	3rd Qu.: 350000
Max. : 860000	Max. : 1000000	Max. : 1200000

5.3 4.3 Numerical analysis

Discuss histograms and boxplots

– EV growth – Using 3 D map

- Germany (1.9M in 2024), France (1.1M), and UK (1.4M) evidently the leader in the market.
- There is an uneven but accelerating electrification.
- After 2020, the growth has more rapid.

– EV growth – Using line scatter plot

Steady Early Growth (2015–2019): Most countries show slow increases in EV metrics, reflecting the initial adoption phase.

Rapid Expansion (2020–2024): A sharp upward trend begins around 2020.

Leading Countries:

Germany shows the steepest rise, indicating major investments and market penetration.

United Kingdom, France, and the Netherlands also display strong growth trajectories.

– EV growth – Using Box plot

The data reveal a polarized EV landscape in Europe, where a few leading nations account for most of the EV stock, while others are still in the early adoption phase.

Germany and France have the highest median EV stock and widest spread, showing both strong adoption and high variability over time — reflecting their large markets and accelerating EV growth.

Norway, Netherlands, and the United Kingdom also exhibit high medians, confirming their established EV infrastructures and mature markets.

Smaller countries like Greece, Iceland, and Poland show very low medians and compressed ranges, indicating slower EV uptake.

Outliers appear in countries such as Portugal, Denmark, and the UK, representing years of exceptionally high growth relative to previous periods.

The overall dispersion suggests significant regional disparity: Western and Northern Europe dominate EV adoption, while Southern and Eastern regions lag behind.

Statistical studies summary : Preliminary study show Strong Positive Correlation between number of electric vehicles (EV stock) and the number of charging station in germany and France.

The data points show a clear upward trend, indicating a strong positive relationship between the number of electric vehicles (EV stock) and the number of charging stations.

There are other parameter than will be discussed more in more final report. This only thr shows the preliminary finding of ev growth in europe.

▼ Show code

```
# Step 4.3.1 : 3D Visualization using Plotly
library(plotly)

#Aggregate data to reduce clutter (sum by year & country)
ev_agg <- filtered_ev_data %>%
  group_by(region_country, year) %>%
  summarise(Total_Stock = sum(value, na.rm = TRUE)) %>%
  ungroup()

# 3D Scatter Plot
```

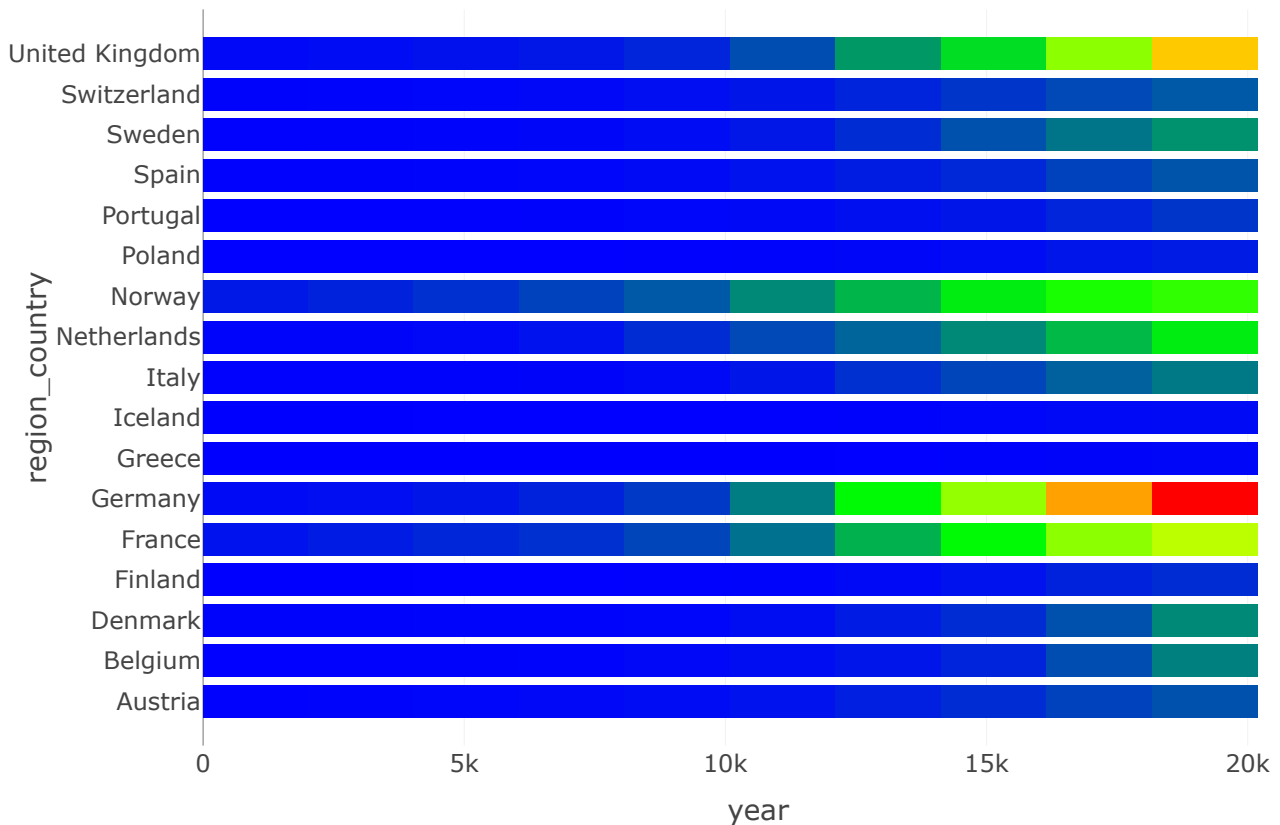
```

plot_ly(
  ev_agg,
  x = ~year,                # X-axis: Time
  y = ~region_country,      # Y-axis: Country names
  z = ~Total_Stock,         # Z-axis: EV stock
  type = "bar",

  color = ~Total_Stock,     # Color by EV stock intensity
  colors = colorRamp(c("blue", "green", "yellow", "red"))
) %>%
layout(
  title = "3D Visualization of BEV Stock Growth in Europe (2015–2024)",
  scene = list(
    xaxis = list(title = "Year"),
    yaxis = list(title = "Country"),
    zaxis = list(title = "EV Stock (Units)")
  )
)

```

3D Visualization of BEV Stock Growth in Europe (2015–2024)



▼ Show code

```

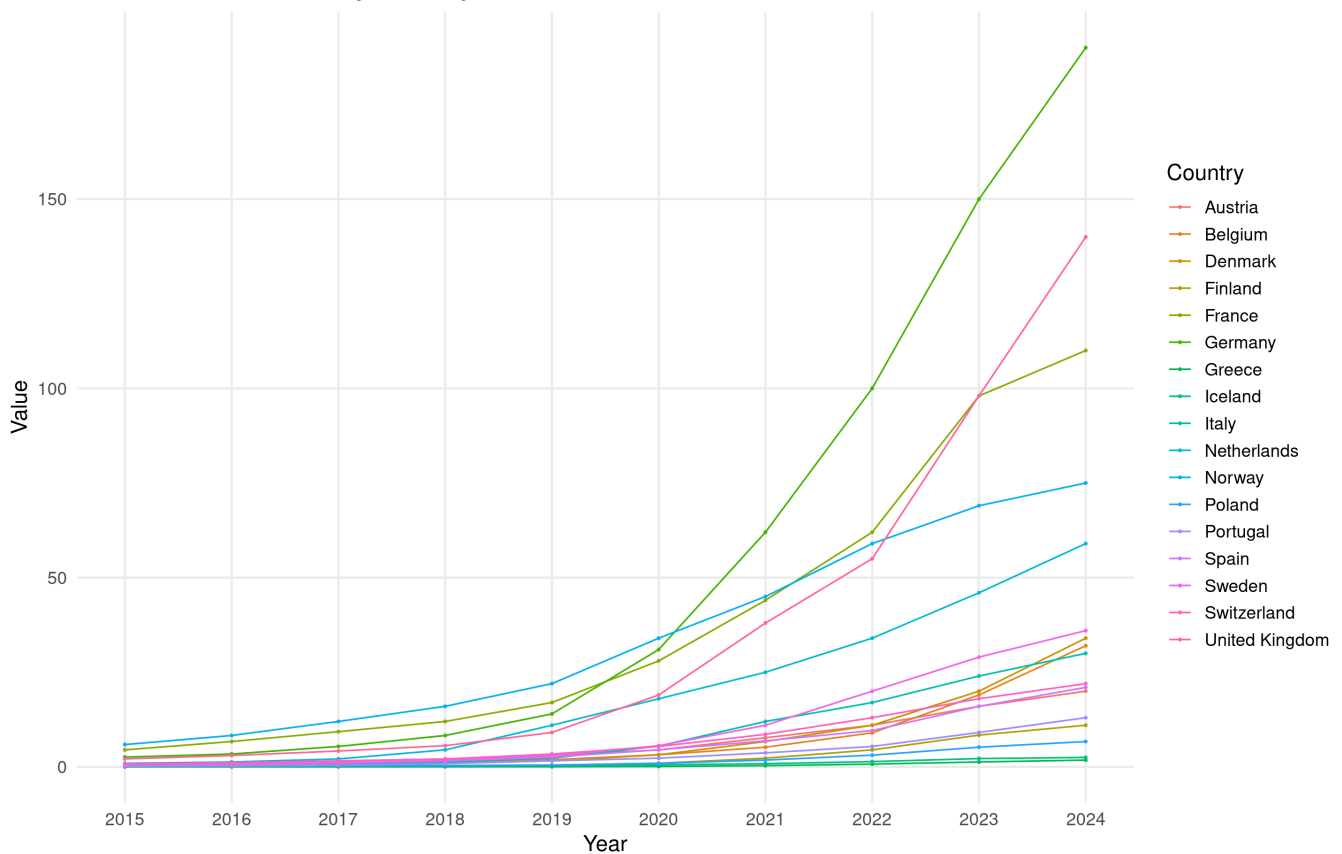
# Step 4.3.2 : Line charts
ev_long <- ev_wide %>%
  pivot_longer(
    cols = -region_country,
    names_to = "year",
    values_to = "value"
  ) %>%
  mutate(year = as.numeric(year))

# --- Plot values as function of time ---

```



```
ggplot(ev_long, aes(x = year, y = value/10000, color = region_country, group = region_
  geom_line(size = 0.5) +
  geom_point(size = 0.5) +
  scale_x_continuous(breaks = c(min(ev_long$year):max(ev_long$year))) +
  theme_minimal(base_size = 14) +
  labs(
    title = "EV Metrics Over Time by Country",
    x = "Year",
    y = "Value",
    color = "Country"
  ) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank()
  )
)
```

EV Metrics Over Time by Country

▼ Show code

```
#-----#
ggplot(ev_long, aes(x = year, y = value, color = region_country, group = region_countr
  geom_line(size = 0.5) +
  geom_point(size = 0.5) +
  scale_y_log10() +
  scale_x_continuous(breaks = c(min(ev_long$year):max(ev_long$year))) +
  theme_minimal(base_size = 14) +
  labs(
    title = "EV Growth Over Time by Country (Log Scale)",
    subtitle = "Visualizing exponential growth trends in EV adoption",

```

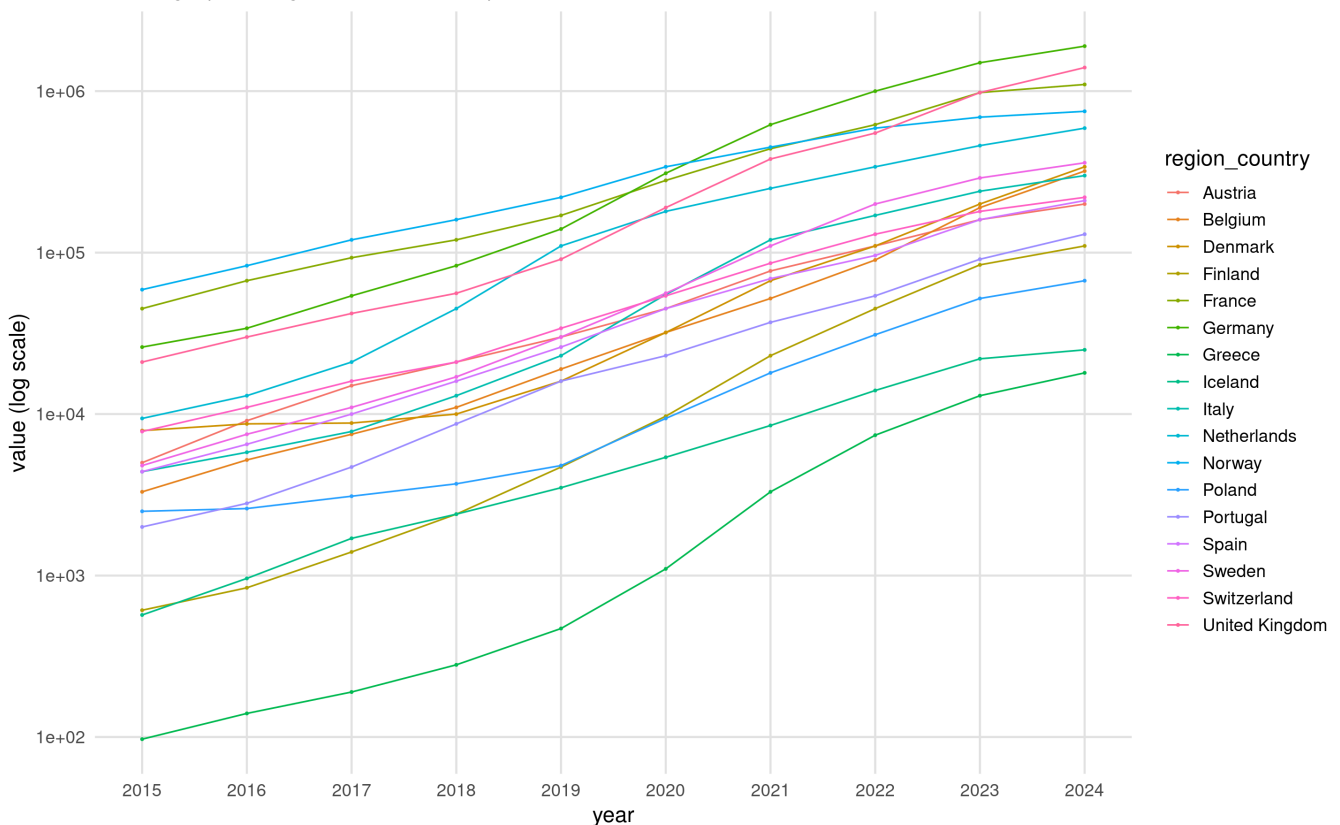
```

  x = "year",
  y = "value (log scale)",
  color = "region_country"
) +
theme(
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "grey90")
)

```

EV Growth Over Time by Country (Log Scale)

Visualizing exponential growth trends in EV adoption



▼ Show code

```

#-----#
ec_long <- ec_wide %>%
  pivot_longer(
    cols = -region_country,
    names_to = "year",
    values_to = "value"
  ) %>%
  mutate(year = as.numeric(year))

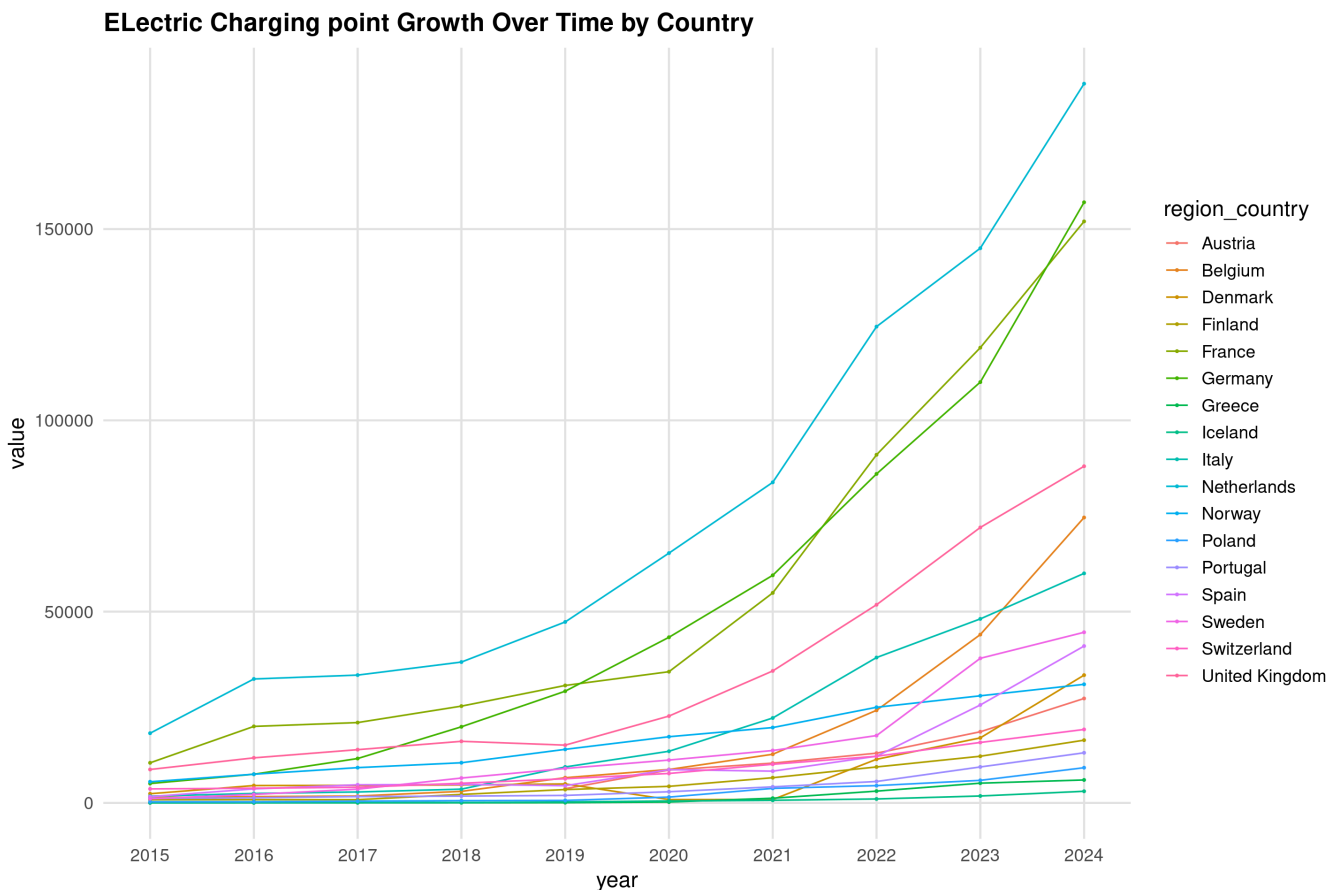
ggplot(ec_long, aes(x = year, y = value, color = region_country, group = region_country)) +
  geom_line(size = 0.5) +
  geom_point(size = 0.5) +
  scale_x_continuous(breaks = c(min(ec_long$year):max(ec_long$year))) +
  theme_minimal(base_size = 14) +
  labs(

```

```

title = "Electric Charging point Growth Over Time by Country",
x = "year",
y = "value",
color = "region_country"
) +
theme(
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "grey90")
)

```



▼ Show code

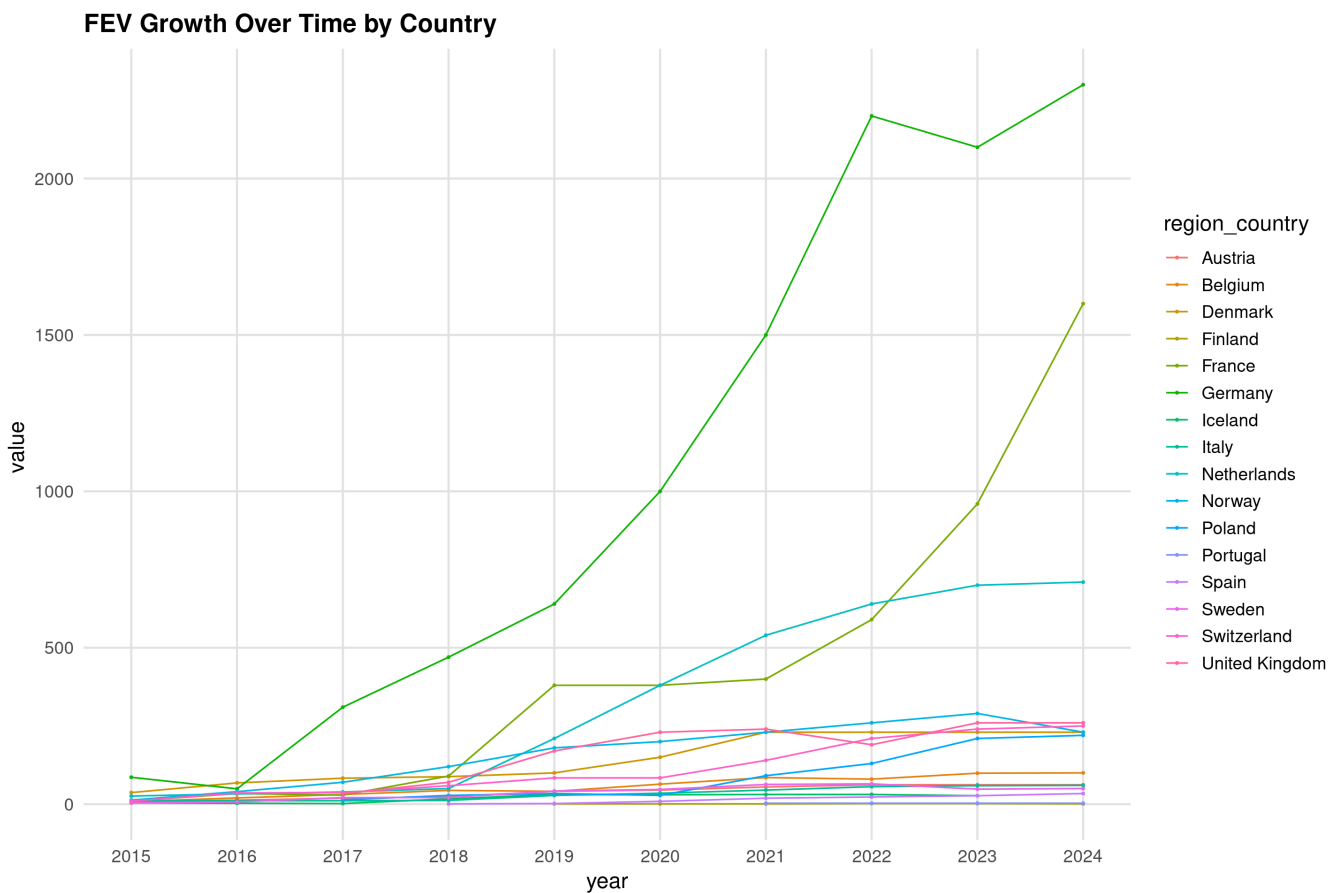
```

#-----#
fev_long <- fev_wide %>%
pivot_longer(
  cols = -region_country,
  names_to = "year",
  values_to = "value"
) %>%
mutate(year = as.numeric(year))

ggplot(fev_long, aes(x = year, y = value, color = region_country, group = region_count
geom_line(size = 0.5) +
geom_point(size = 0.5) +
scale_x_continuous(breaks = c(min(fev_long$year):max(fev_long$year))) +
theme_minimal(base_size = 14) +

```

```
labs(
  title = "FEV Growth Over Time by Country ",
  x = "year",
  y = "value ",
  color = "region_country"
) +
theme(
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "grey90")
)
```



▼ Show code

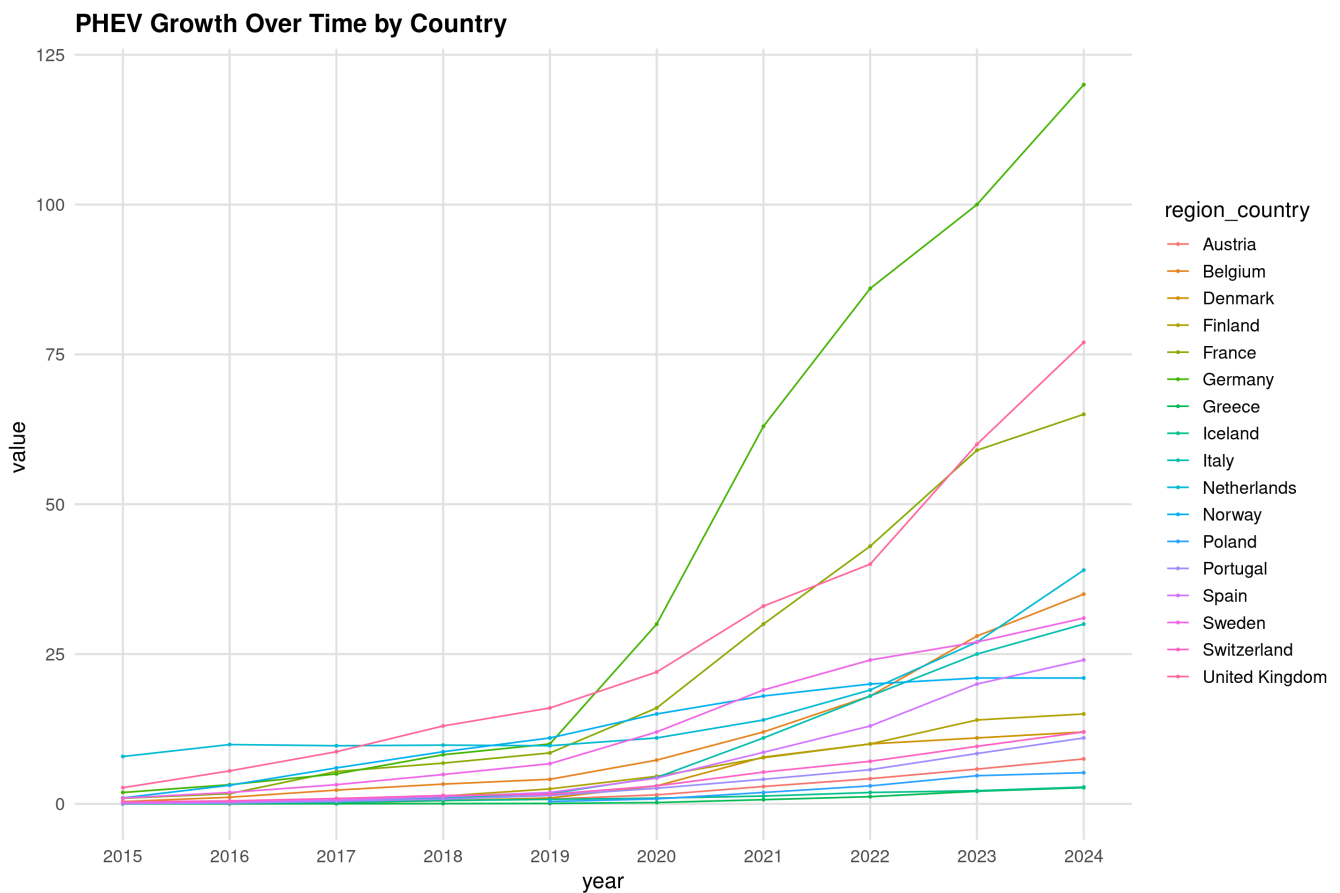
```
#-----#
phev_long <- phev_wide %>%
pivot_longer(
  cols = -region_country,
  names_to = "year",
  values_to = "value"
) %>%
mutate(year = as.numeric(year))

ggplot(phev_long, aes(x = year, y = value/10000, color = region_country, group = regio
geom_line(size = 0.5) +
geom_point(size = 0.5) +
scale_x_continuous(breaks = c(min(phev_long$year):max(phev_long$year))) +
```

```

theme_minimal(base_size = 14) +
labs(
  title = "PHEV Growth Over Time by Country",
  x = "year",
  y = "value ",
  color = "region_country"
) +
theme(
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "grey90")
)

```



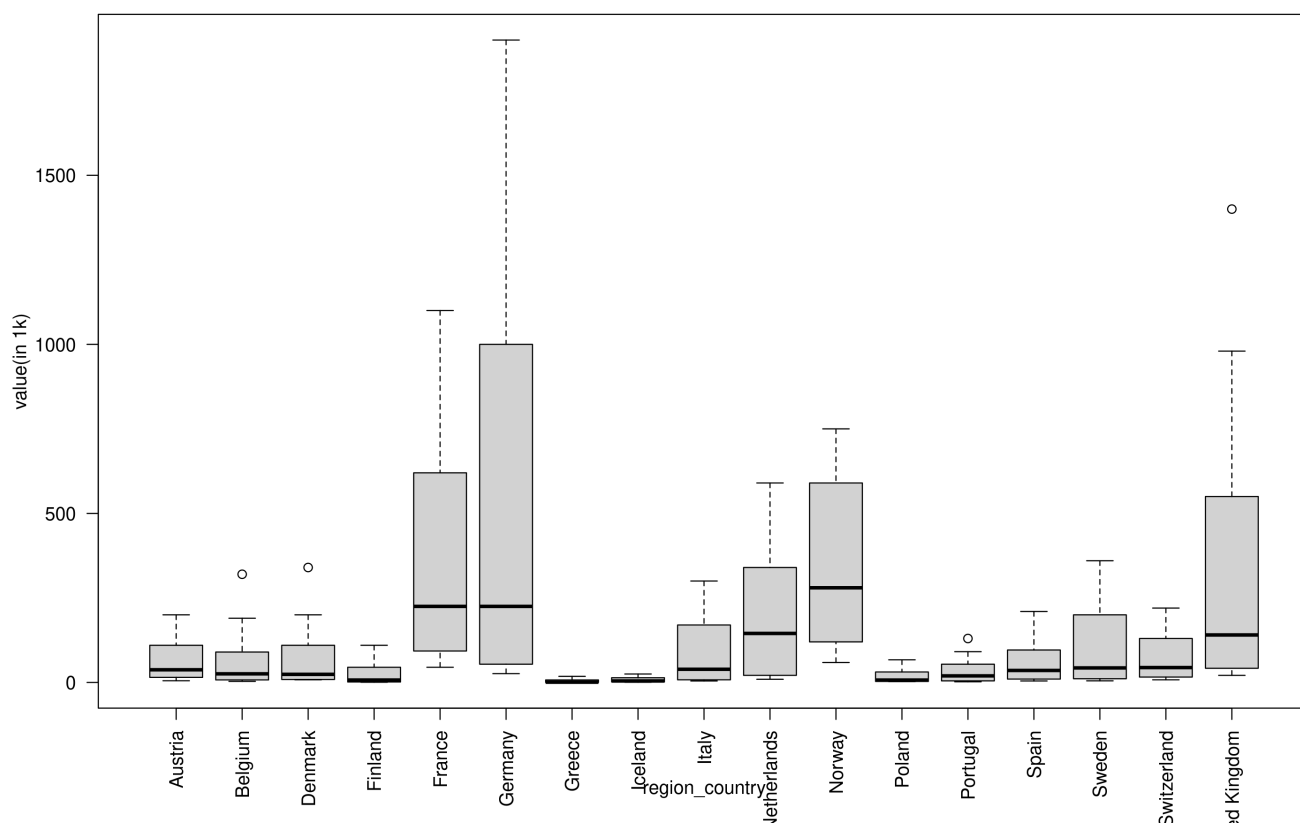
▼ Show code

```

# Step 4.3.3 : Box plot using ggplot
boxplot(
  value/1000 ~ region_country,
  data = ev_long,
  main = "EV Stock in different by Region",
  xlab = "region_country", ylab = "value(in 1k)",
  las = 2 # rotate x labels if long
)

```

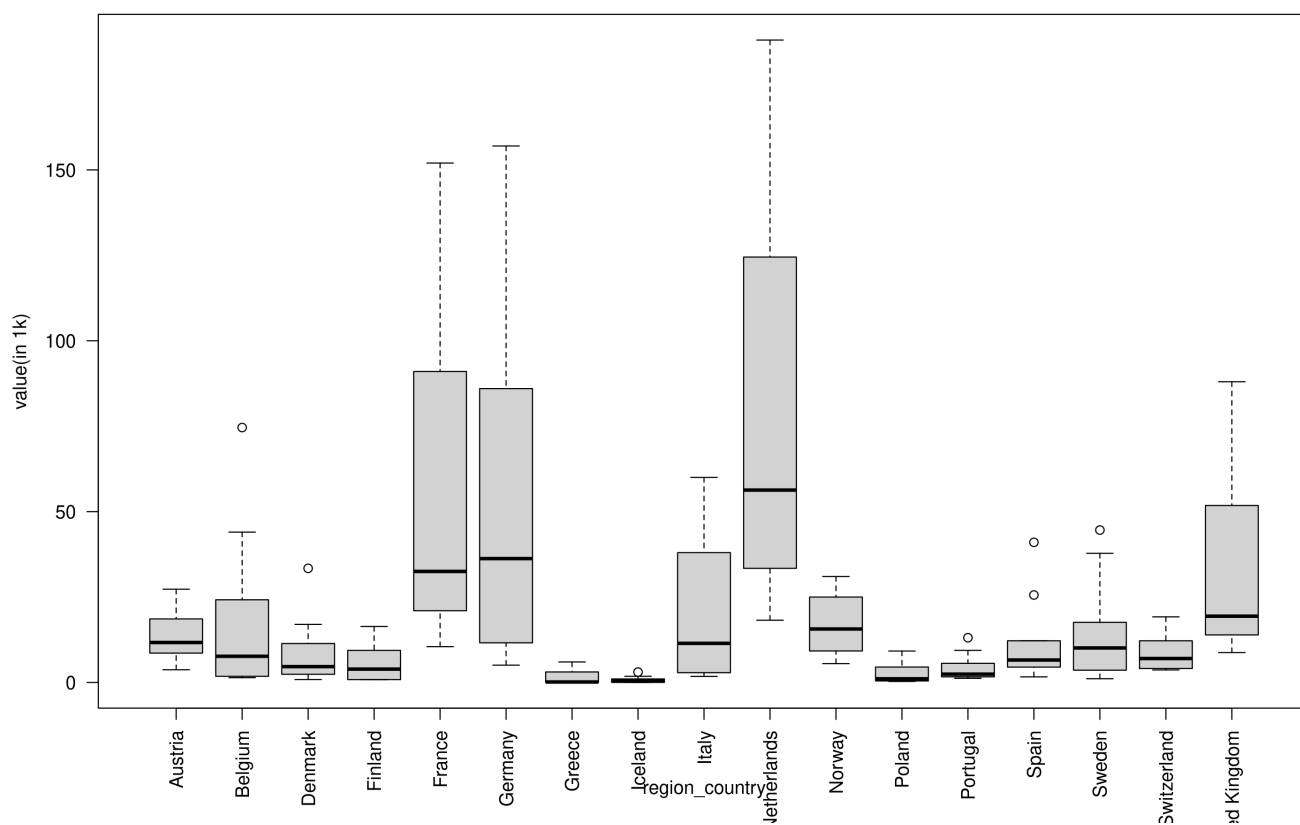
EV Stock in different by Region



▼ Show code

```
boxplot(  
  value/1000 ~ region_country,  
  data = ec_long,  
  main = "Electric charging points in different by Region",  
  xlab = "region_country", ylab = "value(in 1k)",  
  las = 2 # rotate x labels if long  
)
```

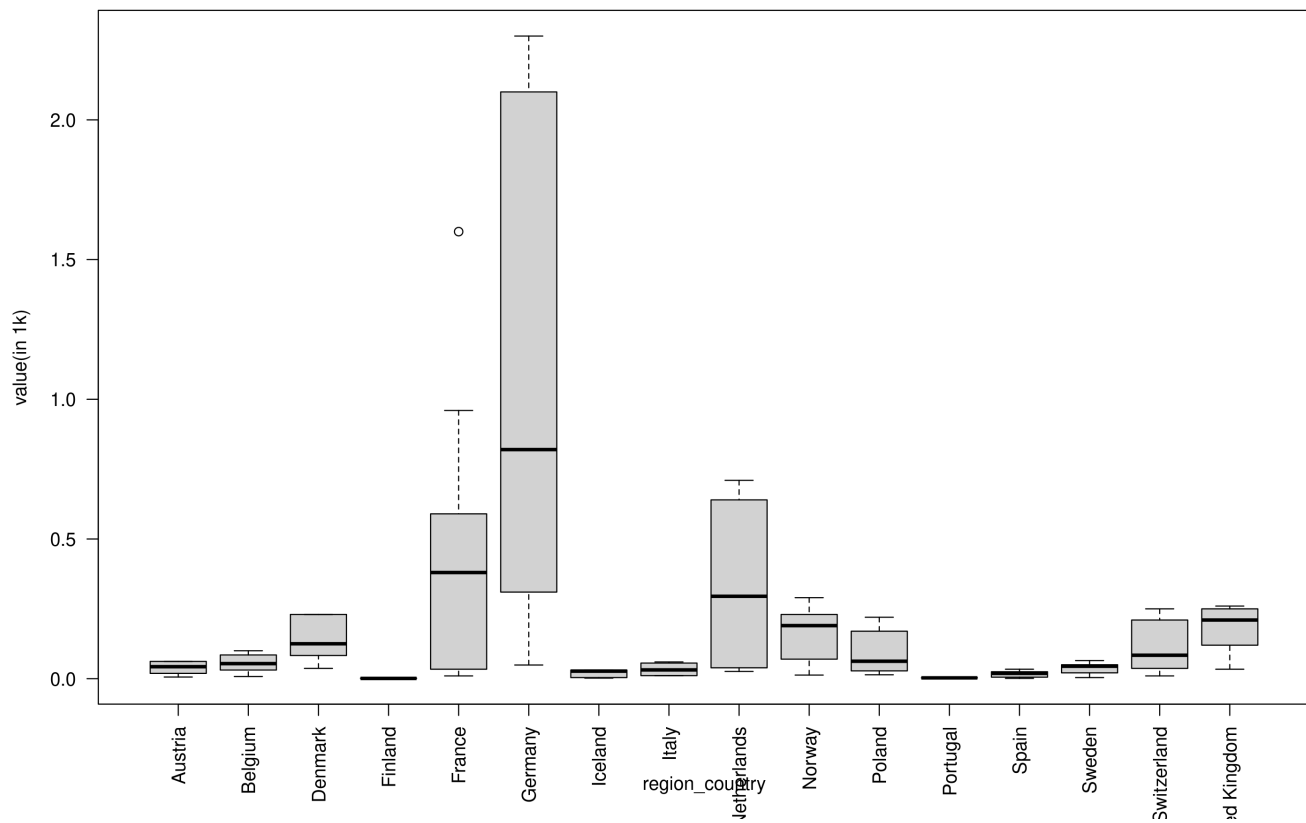
Electric charging points in different by Region



▼ Show code

```
boxplot(  
  value/1000 ~ region_country,  
  data = fev_long,  
  main = "FEV Stock in different by Region",  
  xlab = "region_country", ylab = "value(in 1k)",  
  las = 2 # rotate x labels if long  
)
```

FEV Stock in different by Region



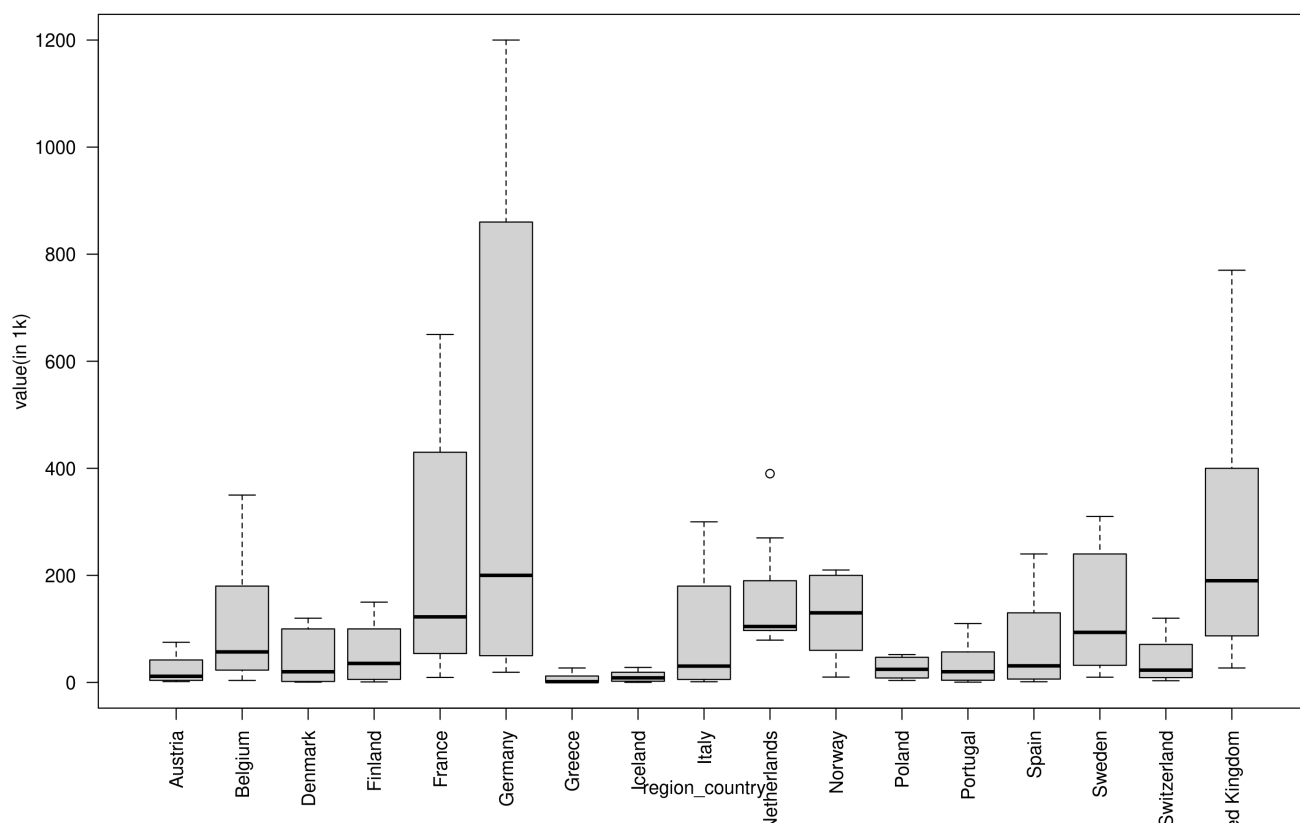
▼ Show code

```

boxplot(
  value/1000 ~ region_country,
  data = phev_long,
  main = "PHEV Stock in different by Region",
  xlab = "region_country", ylab = "value(in 1k)",
  las = 2 # rotate x labels if long
)

```


PHEV Stock in different by Region



5.4 4.4 Categorical analysis

Discuss frequency distributions or group counts.

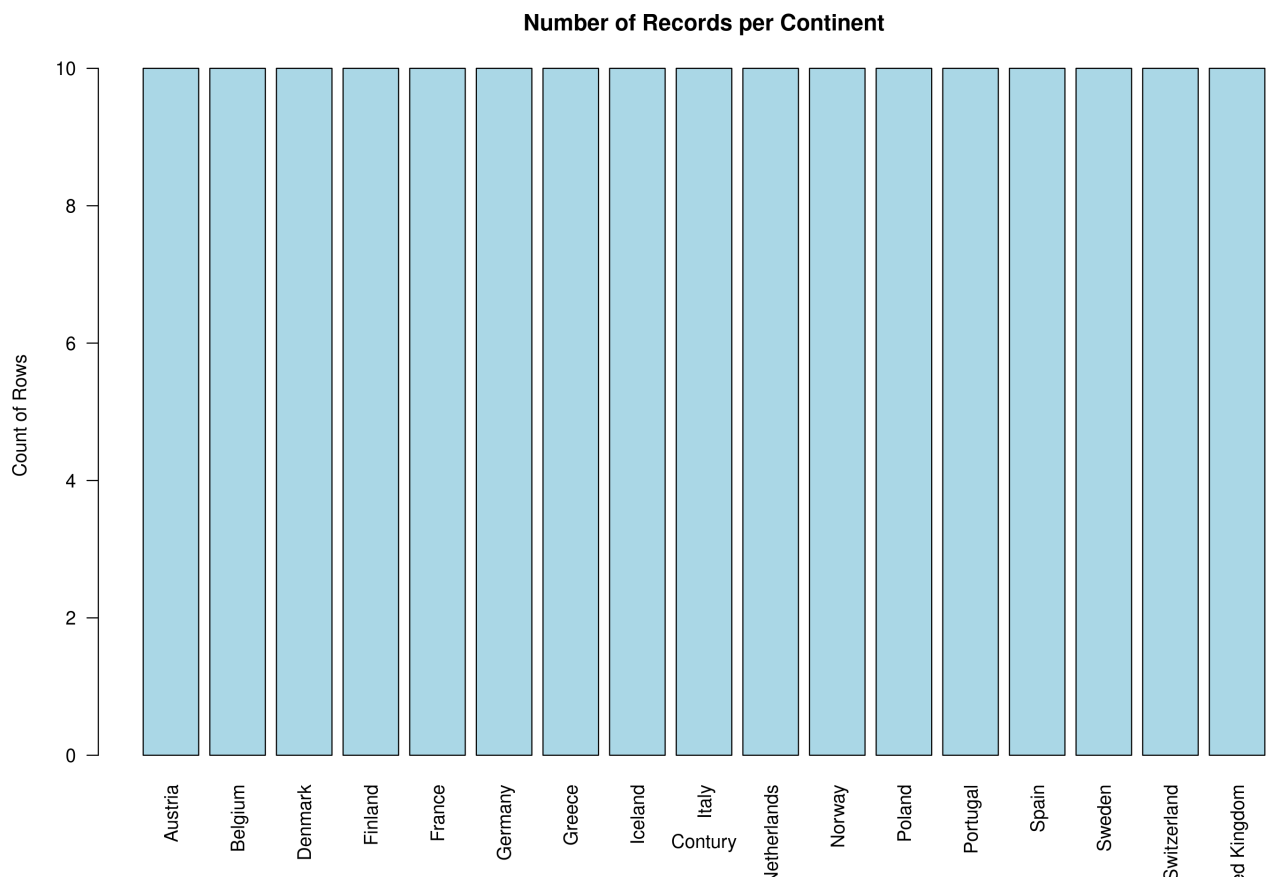
(Insert code cells – counts or plots)

▼ Show code

```
# Step 4.4 : categorical analysis

# Bar plot: counts per continent
continent_counts <- table(ev_long$region_country)

barplot(
  continent_counts,
  main = "Number of Records per Continent",
  xlab = "Contury", ylab = "Count of Rows",
  col = "lightblue",
  las = 2 # rotate x labels
)
```



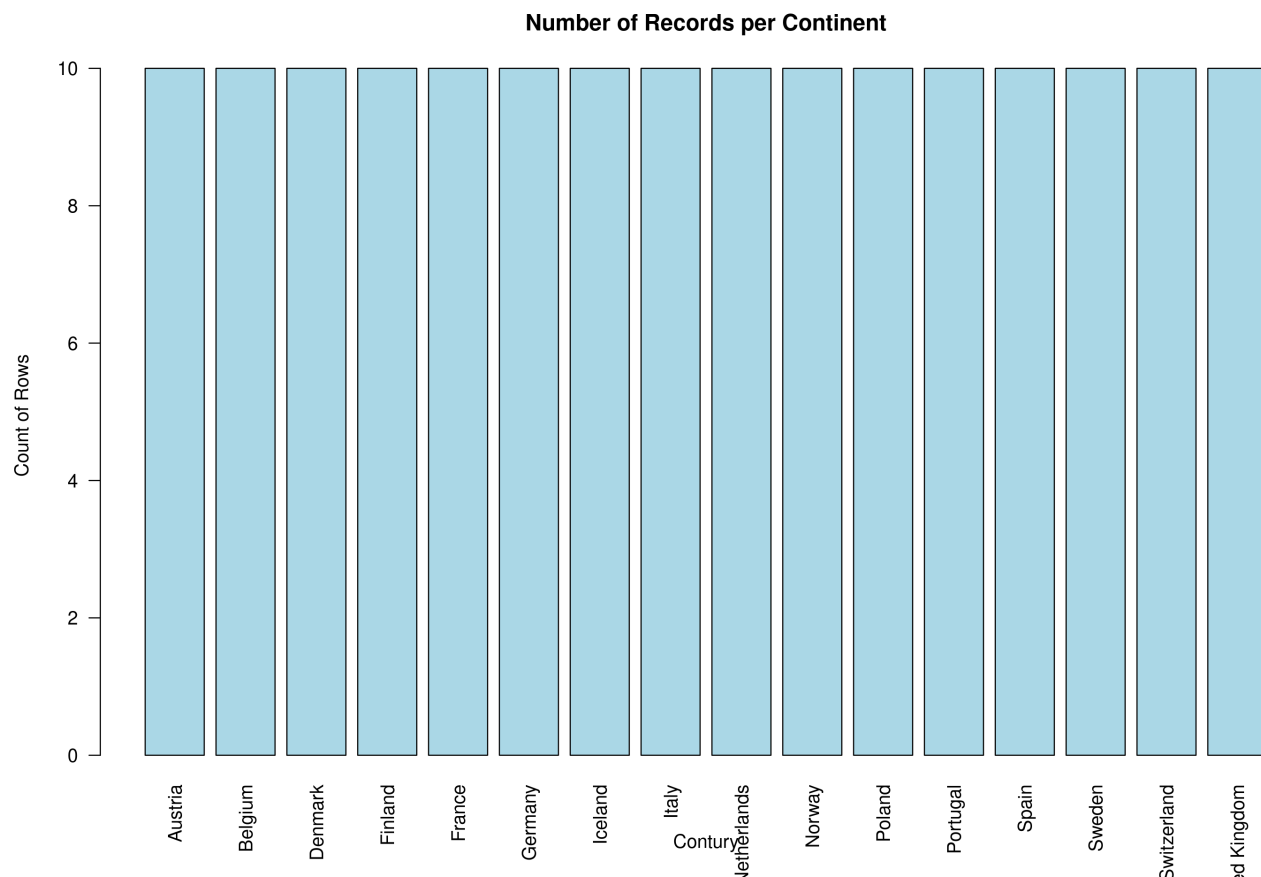
▼ Show code

```
continent_counts1 <- table(ec_long$region_country)

barplot(
  continent_counts1,
  main = "Number of Records per Continent",
  xlab = "Contury", ylab = "Count of Rows",
  col = "lightblue",
  las = 2 # rotate x labels
)

continent_counts2 <- table(fev_long$region_country)
barplot(
  continent_counts1,
  main = "Number of Records per Continent",
  xlab = "Contury", ylab = "Count of Rows",
  col = "lightblue",
  las = 2 # rotate x labels
)

continent_counts3 <- table(phev_long$region_country)
barplot(
  continent_counts2,
  main = "Number of Records per Continent",
  xlab = "Contury", ylab = "Count of Rows",
  col = "lightblue",
  las = 2 # rotate x labels
)
```



▼ Show code

```
# step 4.3.4 : Statistical analysis
top_5 <- c("Germany","France", "United Kingdom")
scatter_dat1 <- ev_long %>%
  filter (region_country == "Germany")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat2 <- ec_long %>%
  filter(region_country == "Germany") %>%
  select(region_country, year, value) %>%
  arrange(desc(year))

germany_data <- merge(scatter_dat1, scatter_dat2, by=c("year","region_country"))

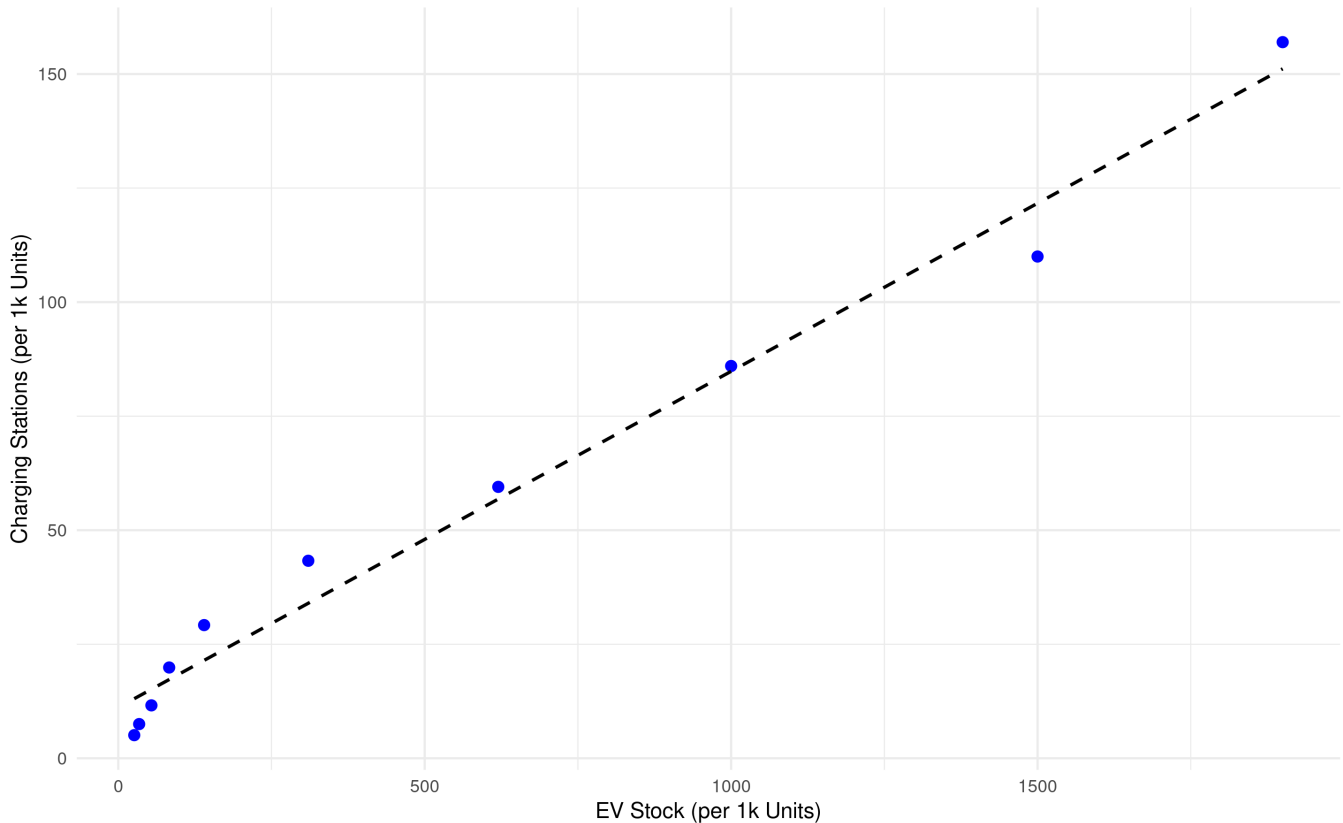
# Scatter Plot: EV stock vs Charging Stations

ggplot(germany_data, aes(x = value.x/1000, y = value.y/1000)) +
  geom_point(aes(), size = 3,color = "blue") +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
  theme_minimal(base_size = 14) +
  labs(
    title = "EV Stock vs. Charging Infrastructure in Germany (2015–2024)",
    subtitle = "Assessing the relationship between vehicle adoption and charging expansion",
    x = "EV Stock (per 1k Units)",
    y = "Charging Stations (per 1k Units)",
  ) +
```

```
theme(
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  legend.position = "right"
)
```

EV Stock vs. Charging Infrastructure in Germany (2015–2024)

Assessing the relationship between vehicle adoption and charging expansion



▼ Show code

```
#-----
scatter_dat3 <- ev_long %>%
  filter (region_country == "France") %>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat4 <- ec_long %>%
  filter(region_country == "France") %>%
  select(region_country, year, value) %>%
  arrange (desc(year))

France_data <- merge(scatter_dat3, scatter_dat4, by=c("year","region_country"))

# Scatter Plot: EV stock vs Charging Stations

ggplot(France_data, aes(x = value.x/1000, y = value.y/1000)) +
  geom_point(aes(), size = 3,color = "blue") +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
  theme_minimal(base_size = 14) +
  labs(
```

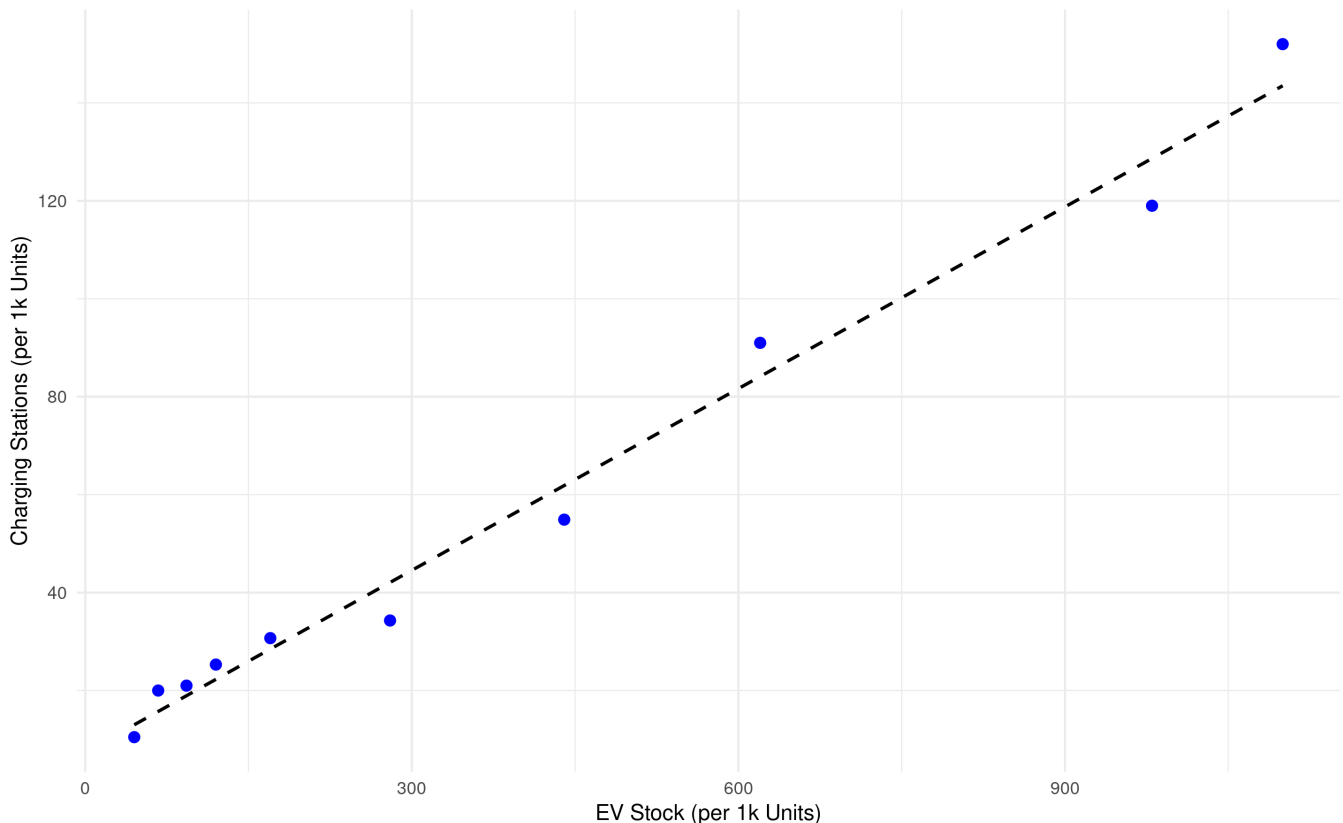
```

title = "EV Stock vs. Charging Infrastructure in France (2015–2024)",
subtitle = "Assessing the relationship between vehicle adoption and charging expansion",
x = "EV Stock (per 1k Units)",
y = "Charging Stations (per 1k Units)",
) +
theme(
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12),
  legend.position = "right"
)

```

EV Stock vs. Charging Infrastructure in France (2015–2024)

Assessing the relationship between vehicle adoption and charging expansion



5.5 4.5 Interpretation

Write short commentary linking patterns to your project context.

The general overall commentary from the data is that, the EV Growth is widespread which is not just a few countries — meaning EV adoption is accelerating across the region. EV adoption became more unequal across Europe over time — early adopters (e.g., Germany, France, Netherlands) surged ahead, while others lagged. Outliers correspond to leaders in EV adoption, not anomalies. They show where charging infrastructure are strongest. The data reflect a clear acceleration in EV adoption across Europe post-2020, with Germany leading the transformation, highlighting the widening gap between high- and low-adoption countries in the EU EV market transition. The overall dispersion suggests significant regional disparity: Western and Northern Europe dominate EV adoption, while Southern and Eastern regions lag behind.

In case of PHEV and FCV, The graph highlights diverging adoption speeds, uptake is widespread, a few leading countries drive most of the regional growth, might be because of the policy effectiveness and market maturity

differences within Europe.

In terms of infrastructure, charging stations, Western European countries (Netherlands, France, Germany, UK) are leading in terms of EV charging infrastructure. Nordic countries like Norway and Sweden show decent deployment but not as high in raw numbers—possibly due to smaller populations. Southern and Eastern Europe (Greece, Poland, Iceland) lag behind, likely reflecting slower adoption or smaller urban infrastructure.

Preliminary study show Strong Positive Correlation between number of electric vehicles (EV stock) and the number of charging station in Germany and France but has to extended to see other countries. The data points show a clear upward trend, indicating a strong positive relationship between the number of electric vehicles (EV stock) and the number of charging stations. this prove our H2 hypothesis : H2 Greater density of public charging stations leads to higher adoption.

6 5 Documentation and Reflection

Capture reasoning, challenges, and decisions.

The learning from this week activity is as follows, Initial Date and research - Data has been pushed to the server and been read from the r module - Hypothesis development using literature. - extracig data from source

Learn lot of new data transformation techiques, - pivot wide - to make it useful analysis - filtering and groupin - str detect function - to detect various text for filtering - checks for missing data - check for duplication - Format check in year, countries and value

Data visualizations - 3d plot - for 3d data visualization - line plot

- To manipulate the legends and axis - box plots - using ggplot - categorical analysis - for char data to check no of entries for all countries - Statistical correlation - linear regression between variable (to be studied more)

Challenges and decisions :

Biggest challenge is connection between R and github. The server is not working from Friday and was to able to push or pull the data i had in r server.

Initial Date and research - Finding data for the entire Europe was a challenge - we have other parameter like population, income levels and fuel prices and all are from different data sets. So per your suggestion, we have started with one data variable of charging point deployment and extend to other.

Learn lot of new data transformation techniques - Using new techniques to filter data was not challenging, without knowing the name but chatgpt was helpful in telling us the required functions. - Data validation is has to be done. Need to form rules for each data set.

Data visualizations

- had to learnt all the data visualization techniques, to study in different way a single variable sto gt new insights
- 3-d visualization was a challenge, because of the complexity of the data , ended up using color graph for 3rd dimension.
- Axis manipulation was a challenge to standardize for each graph

6.1 5.1 What assumptions did you make?

Assumption 1: Cars are only contributing factor for EV growth. Assumption 2: BEV is more significant than FCEV and PHEV from the data Assumption 3 : Data from 2015 to 2024 is only considered for study as it show a stronger growth in most of the countries Assumption 4 : Projection data has not been taken into consideration as it is projection and a actual data set Assumption 5 : Fast charging and slow charging station has been combined and no correlation between found for those separately.

6.2 5.2 What was easy or hard?in

Easy : - Data extraction from website to R - Data transformation through filtering techquies - Stiaistical anyalsis - Writing data to different folders

Hard : Biggest challenge is connection between R and github. The server is not working from friday and was to able to push or pull the data i had in r server.

- 3d Data visualization
- Pivoting
- Graph axis - manipulations

6.3 5.3 What would you do differently next time?

- Has to set up R in my system and link it to the github.
- Analysis of less data, and focus on reflections

7 6 Good Practice Reminders

Checklist and reproducibility notes.

- Comments in code - yes , done
- Relative paths - done and stored in appropriate paths
- Push and pull discipline. - Not able to do it , from my system.
- Consistent naming conventions - yes, as per Euro stat convection

8 7 Group Notes and Variations

Describe context-specific adaptations or creative extensions.

Data visualizations - 3-d visualization was a creative extensions, because of the complexity of the data, ended up using color graph for 3rd dimension.

- Box Plot to study the showing both strong adoption and high variability over time.

Next steps: - using statistical analysis, found relation between the two variables and found to be good correlation with linear.

9 8 Final Checkpoint Before Submission

Tick items off to ensure readiness.

- Dataset in /data/raw/. ok
- Cleaned dataset in /data/processed/ ok
- HTML output in /outputs/ ok
- Reflection written ok
- Changes committed and pushed NO

Week 3 – Understanding Your Data: Provenance, Quality & Risks

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PUBLISHED
November 17, 2025

Overview

What this file is for

This week’s focus is more on **reflection and documentation**, and less or not on coding.
You will describe where your data comes from, how trustworthy it is, and what risks or limitations it may carry.

1 Dataset Overview

A short paragraph (3–5 sentences) describing:

- What your dataset is about
- Who published it
- When and how you accessed it
- Why it fits your project idea

The obtained data includes country-level which includes EU27 data on EV sales(BEV, PHEV, FCEV) and charging point deployment. The data is published by IEA (international Energy Agency) Global EV Policy Explorer. The last version was published on july 2025 and then it was downloaded directly from <https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer> on oct 15th. The key objective of this study is try to quantify the influence of demographic, economic, and infrastructural variables on EV adoption rates.The main hypothesis of our study are, H1 Higher median income positively correlates with EV adoption. H2 Greater density of public charging stations leads to higher adoption. H3 Regions with higher fuel prices have faster EV uptake. And this data helps us make a preliminary analysis of the H2 hypothesis

2 Provenance Summary

Use the table below to record evidence and assessments about your data set. You can edit or expand it as needed. Please provide short support statement (1-2 lines each) for your verdicts.

Criterion	Evidence Checked	Verdict
Origin & Authority	IEA Global has created it from various sources and published it. Purpose: The Global EV Outlook is an annual publication that identifies and discusses recent developments in electric mobility across the globe. IEA analysis based on EV Volumes,	

Criterion	Evidence Checked	Verdict
	Marklines, ACEA, EFAO, OICA data. With this data, it provides recommendations from leading markets to inform policy makers and stakeholders with regard to policy frameworks and market systems for electric vehicle adoption.	
Coverage & Completeness	The data cover from 2010 to 2024 with projections upto 2030 for worldwide countries. It has many variables, with main focus electric vehicle sales and charging infrastructure deployment, energy use, CO2 emissions, battery demand and related policy developments. The sample size is 16436 observations. We have “Na” in many fields, but these are not missing data, but not yet available in those countries.	✓
Timeliness & Stability	Last update on 31st July’25. The Data is updated annually in July to reflect last year’s EV global trends. The schema is consistent with variables and structure, but new indicators are added (e.g. charging point type) are added on every update.	✓
Format & Access	The data was extracted in excel format, but was available in both xlxs and csv format. The data could not be accessed through import path because the data does not have separate weblink.	✗
Documentation & Versioning	The data dictionary is not available for the data. The change log is available in the readme file in the excel file. Version number is available for this data and it is version 1.2	⚠

Criterion	Evidence Checked	Verdict
License & Attribution	The license for this data is CC by 4.0 License (Creative Commons (CC) licenses). The url is https://www.iea.org/terms/creative-commons-cc-licenses. The IEA's preferred attribution is: IEA 202[X]; [Title name], [URL], License: [relevant CC license e.g. CC BY 4.0];.	✓
Sensitivity & Ethics	Low risk. There is no sensitivity information or ethical data with no personally identifiable information in the dataset. It contains openly published data for different countries all over the world	✓
Data Collection method	The data is aggregated from various national reports, industry data, and IEA modeling frameworks such as the Global EV Outlook and STEPS scenario.	✓
Quality & Verification	The data has been validated against national statistics, manufacturer reports, and energy system models to ensure consistency.	✓

2.1 Understanding Each Criterion

- **Origin & Authority:** Identify who created or published the dataset, and why it exists. Check for official sources or credible research institutions.
- **Coverage & Completeness:** Consider whether the data spans a sufficient period, geography, and number of observations for your analysis. Missing years or regions may signal gaps.
- **Timeliness & Stability:** When was it last updated? Has the structure changed over time? Outdated or unstable sources may reduce reliability.
- **Format & Access:** Ensure the format (CSV, JSON, etc.) can be opened reproducibly with R/Python. Record any URLs or APIs used.
- **Documentation & Versioning:** Look for accompanying documentation (README, codebook, changelog) or version numbers that clarify structure.
- **License & Attribution:** Record the license name (e.g., CC BY 4.0, OGL v3.0) and link. Copy the required attribution statement if available.

- **Sensitivity & Ethics:** Note whether the data includes personal, identifiable, or ethically sensitive information. If yes, explain how you'll handle it responsibly.

2.2 Licensing Details

Fill in the fields below using information from your dataset's source page. Copy the exact text if provided.

Criterion	Information
License Name:	Creative Commons Attribution 4.0 license (CC BY 4.0).
License Link:	https://www.iea.org/terms/creative-commons-cc-licenses .
Attribution Text (if provided):	The IEA's preferred attribution is: IEA 202[X]; [Title name], [URL], License: [relevant CC license e.g. CC BY 4.0];
Notes on Limitations (e.g., Non-Commercial):	Licensing: It allows sharing and adaptation of the data for any purpose, including commercial use, provided appropriate credit is given to the IEA. However, derivative works must not imply IEA endorsement or representation. Schema Evolution: There are periodic updates with new variable names, structure can cause schema inconsistency across years.

Tip

If you cannot find a license, state:

"No explicit license visible; flagged for clarification or restricted use."

3 Early Risks Reflection

Write about **150 words** identifying key risks or concerns, such as:

THE IEA EV data set has a wide range of information spanning different countries across the globe which can be used for EV trend analysis, but it has its fair share of risks and are listed below.

Key risks identified :

- The are missing and incomplete data for some countries especially smaller ones, particularly before 2017, leading to uneven temporal coverage.
- Inconsistencies in data collection methodology between countries can affect comparability. As data is collected from national agencies who reply on on self-reported data from manufacturers or registration authorities with varying accuracy standards, we have risk of report bias.

- the licensing and provenance limitations under the Creative Commons Attribution 4.0 license impose constraints on data redistribution and modification.

These can affect our correlation studies, as we plan to rank the parameters in the level of importance that influence the EV sales and these biases can bring geographical discrepancies for ex, top 5 EV sales EU nations can behave differently compared to others.

To mitigate these, data validation checks has to be performed in identifying missing or implausible values, comparing aggregated totals against official IEA statistics, and ensuring consistent units of measurement. Outliers should be examined to distinguish between genuine market shifts and data entry errors.

Tip: Think “If someone reused our results, what could mislead them?”

4 (Optional) API Comparison

If applicable, you may compare data access via direct download vs API.

Below is an optional R code snippet to illustrate how APIs differ from static downloads.

▼ Show code

```
library(httr)
library(tidyverse)
library(readxl)
library(jsonlite)
library(readr)

# Download CSV (static)
url1<-'https://www.iea.org/product/download/020887-000391-020756'
p1f <- tempfile()
download.file(url1, p1f, mode="wb")
p1<-read_excel(path = p1f, sheet = 1)
#XL_data <- read_excel("https://www.iea.org/product/download/020887-000391-020756.xlsx")

# Access via API (dynamic)
response <- GET("https://www.iea.org/product/download/020887-000391-020756")
api_data <- fromJSON(content(response, "text"))

# Compare basic structures
names(csv_data)
names(api_data)
```

Summarise your observations (e.g., freshness, column structure, metadata differences).

The API release for the data is protected by the API key. And the API key is only offered to customers purchasing an Enterprise or Global Enterprise licence. So it was not possible to extract it by API method. evidence:

<https://www.iea.org/documentation>

5 (Optional) Scouting / Alternative Data

List any other data sets you explored (name, link, why rejected or unsuitable). These could be the data sets that are somehow comparable to what you are currently using.

A short bullet list is enough.

Dataset / Source	Relation to Main Dataset	Outcome / Issues
road_eqs_zev source:Eurostat https://doi.org/10.2908/ROAD_EQS_ZEV	EV sales information for complete European countries	Positives : - More accurate information; current data is round to 1000's - Available for all European countries; current date is focused on 12 countries Negatives : Data size is huge Has only information EV sales, need to extract separate data sets for each parameters(Charging point, Income, Population etc..) with different data structures which becomes extremely complicated

Week 4 – Statistics: Distributions & First Model

AUTHOR

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PUBLISHED

November 17, 2025

Overview

What this file is for

This week is hands-on with **statistics**. You will: - Measure spread (**SD**, **IQR**) for two numerical variables - Visualise **distribution shapes** (histograms + Normal overlays) - Fit a **simple linear regression** (e.g., `grade ~ study_hours`) - Inspect **residuals** to check model fit - Record brief **group reflections**

Work with your **cleaned Week 2 dataset** (no impossible or extreme values).

1 Dataset & Setup

Use your cleaned dataset prepared in Week 2. Mind the file path.

▼ Show code

```
library(tidyverse)
library(readxl)
library(dplyr)
options(dplyr.summarise.inform = FALSE)
knitr::opts_chunk$set(fig.width = 12,
                      fig.height = 8,
                      include = TRUE,
                      echo = TRUE)

# Load your cleaned dataset
filtered_ev_data <- read_csv("data/processed/EV_Europe_BEV_Stock_2015_2024.csv")
filtered_ec_data <- read_csv("data/processed/EC_Europe_BEV_Stock_2015_2024.csv")

# Quick structure check
glimpse(filtered_ev_data)
```

Rows: 170

Columns: 3

```
$ region_country <chr> "Germany", "Germany", "United Kingdom", "France", "Germ...
$ year           <dbl> 2024, 2023, 2024, 2024, 2022, 2023, 2023, 2024, 2023, 2...
$ value          <dbl> 1900000, 1500000, 1400000, 1100000, 1000000, 980000, 98...
```

▼ Show code

```
glimpse(filtered_ec_data)
```

Rows: 325

Columns: 3

```
$ region_country <chr> "Netherlands", "Netherlands", "France", "Germany", "Net...
```

```
$ year      <dbl> 2024, 2023, 2024, 2024, 2022, 2023, 2023, 2021, 2022, 2...
$ value     <dbl> 180000, 140000, 120000, 120000, 120000, 99000, 87000, 8...
```

▼ Show code

```
colSums(is.na(filtered_ev_data))
```

```
region_country      year      value
              0              0              0
```

▼ Show code

```
colSums(is.na(filtered_ec_data))
```

```
region_country      year      value
              0              0              0
```

Confirm (1–2 lines): that the dataset loaded correctly (rows/columns) and that `grade` and `study_hours` are numeric.

The processed data sets with two variables i.e. EV sales and Electric charging points for each of the european countries from the year 2015 to 2024 has been extracted and stored in `filtered_ev_data` and `filtered_ec_data` respectively. Both the variables are numeric i.e double class type. The data has been validated for “NA” values and found to be none and also checked for inconsistencies in week2 analysis.

2 Spread & Variability

Compute summaries, SD, and IQRs for at least two two main variables from your dataset (e.g., `grade` and `study_hours`).

▼ Show code

```
# Summaries

scatter_dat_France_ev <- filtered_ev_data%>%
  filter (region_country == "France")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Germany_ev <- filtered_ev_data%>%
  filter (region_country == "Germany")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_uk_ev <- filtered_ev_data%>%
  filter (region_country == "United Kingdom")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Norway_ev <- filtered_ev_data%>%
  filter (region_country == "Norway")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))
```



```
scatter_dat_Netherlands_ev <- filtered_ev_data%>%
  filter (region_country == "Netherlands")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_France_ec <- filtered_ec_data%>%
  filter (region_country == "France")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Germany_ec <- filtered_ec_data%>%
  filter (region_country == "Germany")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_uk_ec <- filtered_ec_data%>%
  filter (region_country == "United Kingdom")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Norway_ec <- filtered_ec_data%>%
  filter (region_country == "Norway")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Netherlands_ec <- filtered_ec_data%>%
  filter (region_country == "Netherlands")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

# summary for EV sales for top 5 countries
summary(scatter_dat_France_ev)
```

region_country	year	value
Length:10	Min. :2015	Min. : 45000
Class :character	1st Qu.:2017	1st Qu.: 99750
Mode :character	Median :2020	Median : 225000
	Mean :2020	Mean : 391500
	3rd Qu.:2022	3rd Qu.: 575000
	Max. :2024	Max. :1100000

► Show code

region_country	year	value
Length:10	Min. :2015	Min. : 26000
Class :character	1st Qu.:2017	1st Qu.: 61250
Mode :character	Median :2020	Median : 225000
	Mean :2020	Mean : 566700
	3rd Qu.:2022	3rd Qu.: 905000
	Max. :2024	Max. :1900000

▼ Show code

```
summary(scatter_dat_uk_ev)
```

region_country	year	value
Length:10	Min. :2015	Min. : 21000
Class :character	1st Qu.:2017	1st Qu.: 45500
Mode :character	Median :2020	Median : 140500
	Mean :2020	Mean : 374000
	3rd Qu.:2022	3rd Qu.: 507500
	Max. :2024	Max. :1400000

▼ Show code

```
summary(scatter_dat_Norway_ev)
```

region_country	year	value
Length:10	Min. :2015	Min. : 59000
Class :character	1st Qu.:2017	1st Qu.:130000
Mode :character	Median :2020	Median :280000
	Mean :2020	Mean :346200
	3rd Qu.:2022	3rd Qu.:555000
	Max. :2024	Max. :750000

▼ Show code

```
summary(scatter_dat_Netherlands_ev)
```

region_country	year	value
Length:10	Min. :2015	Min. : 9400
Class :character	1st Qu.:2017	1st Qu.: 27000
Mode :character	Median :2020	Median :145000
	Mean :2020	Mean :201840
	3rd Qu.:2022	3rd Qu.:317500
	Max. :2024	Max. :590000

▼ Show code

```
# summary for Electric charging points for top 5 countries  
summary(scatter_dat_France_ec)
```

region_country	year	value
Length:20	Min. :2015	Min. : 580
Class :character	1st Qu.:2017	1st Qu.: 3150
Mode :character	Median :2020	Median : 19500
	Mean :2020	Mean : 27934
	3rd Qu.:2022	3rd Qu.: 31250
	Max. :2024	Max. :120000

▼ Show code

```
summary(scatter_dat_Germany_ec)
```

region_country	year	value
Length:20	Min. :2015	Min. : 470
Class :character	1st Qu.:2017	1st Qu.: 4250
Mode :character	Median :2020	Median : 12250
	Mean :2020	Mean : 26454
	3rd Qu.:2022	3rd Qu.: 37000
	Max. :2024	Max. :120000

▼ Show code

```
summary(scatter_dat_uk_ec)
```

region_country	year	value
Length:20	Min. :2015	Min. : 540
Class :character	1st Qu.:2017	1st Qu.: 3800
Mode :character	Median :2020	Median :11500
	Mean :2020	Mean :16732
	3rd Qu.:2022	3rd Qu.:17250
	Max. :2024	Max. :71000

▼ Show code

```
summary(scatter_dat_Norway_ec)
```

region_country	year	value
Length:20	Min. :2015	Min. : 330
Class :character	1st Qu.:2017	1st Qu.: 4900
Mode :character	Median :2020	Median : 8800
	Mean :2020	Mean : 8388
	3rd Qu.:2022	3rd Qu.:12250
	Max. :2024	Max. :18000

▼ Show code

```
summary(scatter_dat_Netherlands_ec)
```

region_country	year	value
Length:20	Min. :2015	Min. : 220
Class :character	1st Qu.:2017	1st Qu.: 2050
Mode :character	Median :2020	Median : 13000
	Mean :2020	Mean : 38738
	3rd Qu.:2022	3rd Qu.: 50250
	Max. :2024	Max. :180000

▼ Show code

```
# # Standard Deviation for EV sales for top 5 countries
# sd(scatter_dat_Germany_ev$value, na.rm = TRUE)
# sd(scatter_dat_France_ev$value, na.rm = TRUE)
# sd(scatter_dat_uk_ev$value, na.rm = TRUE)
# sd(scatter_dat_Norway_ev$value, na.rm = TRUE)
# # Standard Deviation for Electric charging points for top 5 countries
```

```
# sd(scatter_dat_France_ec$value,na.rm = TRUE)
# sd(scatter_dat_Germany_ec$value,na.rm = TRUE)
# sd(scatter_dat_uk_ec$value,na.rm = TRUE)
# sd(scatter_dat_Norway_ec$value,na.rm = TRUE)
# # IQR for EV sales for top 5 countries
# IQR(scatter_dat_Germany_ev$value,na.rm = TRUE)
# IQR(scatter_dat_France_ev$value,na.rm = TRUE)
# IQR(scatter_dat_uk_ev$value,na.rm = TRUE)
# IQR(scatter_dat_Norway_ev$value,na.rm = TRUE)
# # IQR for Electric charging points for top 5 countries
# IQR(scatter_dat_France_ec$value,na.rm = TRUE)
# IQR(scatter_dat_Germany_ec$value,na.rm = TRUE)
# IQR(scatter_dat_uk_ec$value,na.rm = TRUE)
# IQR(scatter_dat_Norway_ec$value,na.rm = TRUE)

summary_Top4_evsales <- tibble(
  Country = c("Germany","France","Uk", "Norway",'Netherlands'),
  Max = c(max(scatter_dat_Germany_ev$value),max(scatter_dat_France_ev$value),max(scatter_dat_uk_ev$value),max(scatter_dat_Norway_ev$value),max(scatter_dat_Netherlands_ev$value)),
  Min = c(min(scatter_dat_Germany_ev$value),min(scatter_dat_France_ev$value),min(scatter_dat_uk_ev$value),min(scatter_dat_Norway_ev$value),min(scatter_dat_Netherlands_ev$value)),
  Mean = c(mean(scatter_dat_Germany_ev$value),mean(scatter_dat_France_ev$value),mean(scatter_dat_uk_ev$value),mean(scatter_dat_Norway_ev$value),mean(scatter_dat_Netherlands_ev$value)),
  Median = c(median(scatter_dat_Germany_ev$value),median(scatter_dat_France_ev$value),median(scatter_dat_uk_ev$value),median(scatter_dat_Norway_ev$value),median(scatter_dat_Netherlands_ev$value)),
  Std_Dev = c(sd(scatter_dat_Germany_ev$value),sd(scatter_dat_France_ev$value),sd(scatter_dat_uk_ev$value),sd(scatter_dat_Norway_ev$value),sd(scatter_dat_Netherlands_ev$value)),
  IQR = c(IQR(scatter_dat_Germany_ev$value),IQR(scatter_dat_France_ev$value),IQR(scatter_dat_uk_ev$value),IQR(scatter_dat_Norway_ev$value),IQR(scatter_dat_Netherlands_ev$value))
summary_Top4_evsales
```

Country <chr>	Max <dbl>	Min <dbl>	Mean <dbl>	Median <dbl>	Std_Dev <dbl>	IQR <dbl>
Germany	1900000	26000	566700	225000	679889.5	843750
France	1100000	45000	391500	225000	387662.6	475250
Uk	1400000	21000	374000	140500	473657.4	462000
Norway	750000	59000	346200	280000	259278.0	425000
Netherlands	590000	9400	201840	145000	204838.6	290500

5 rows

▼ Show code

```
summary_Top4_ec <- tibble(
  Country = c("Germany","France","Uk", "Norway","Netherlands"),
  Max = c(max(scatter_dat_Germany_ec$value),max(scatter_dat_France_ec$value),max(scatter_dat_uk_ec$value),max(scatter_dat_Norway_ec$value),max(scatter_dat_Netherlands_ec$value)),
  Min = c(min(scatter_dat_Germany_ec$value),min(scatter_dat_France_ec$value),min(scatter_dat_uk_ec$value),min(scatter_dat_Norway_ec$value),min(scatter_dat_Netherlands_ec$value)),
  Mean = c(mean(scatter_dat_Germany_ec$value),mean(scatter_dat_France_ec$value),mean(scatter_dat_uk_ec$value),mean(scatter_dat_Norway_ec$value),mean(scatter_dat_Netherlands_ec$value)),
  Median = c(median(scatter_dat_Germany_ec$value),median(scatter_dat_France_ec$value),median(scatter_dat_uk_ec$value),median(scatter_dat_Norway_ec$value),median(scatter_dat_Netherlands_ec$value)),
  Std_Dev = c(sd(scatter_dat_Germany_ec$value),sd(scatter_dat_France_ec$value),sd(scatter_dat_uk_ec$value),sd(scatter_dat_Norway_ec$value),sd(scatter_dat_Netherlands_ec$value)),
  IQR = c(IQR(scatter_dat_Germany_ec$value),IQR(scatter_dat_France_ec$value),IQR(scatter_dat_uk_ec$value),IQR(scatter_dat_Norway_ec$value),IQR(scatter_dat_Netherlands_ec$value))
summary_Top4_ec
```

Country <chr>	Max <dbl>	Min <dbl>	Mean <dbl>	Median <dbl>	Std_Dev <dbl>	IQR <dbl>
Germany	120000	470	26453.5	12250	32663.312	32750
France	120000	580	27934.0	19500	33612.857	28100
Uk	71000	540	16732.0	11500	19830.054	13450

Country	Max	Min	Mean	Median	Std_Dev	IQR
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Norway	18000	330	8387.5	8800	5365.789	7350
Netherlands	180000	220	38737.5	13000	52761.749	48200

5 rows

Comment (2–3 lines): Which variable varies more? Are the ranges reasonable? What does the **IQR** say about the middle 50%?

With the overall EV sales and electrical charging point data, it is difficult to form any inferences as the behavior varies a lot with each country. So the data was extracted for top 5 countries (Germany, France, Uk, Netherlands, Norway) in terms of sales and charging points and analysed.

Variable 1 (EV Sales): Germany has the highest mean, SD and IQR compared to others, which shows very high accelerated growth. The range is seems reasonable for all countries, growth has been exponential. IQR is highest for Germany, but similar for Uk, France and Norway, which means the growth in the middle 50% i.e from year 2018 to 2021 is same for these countries. But if we compare their SD, UK has better SD than other two, which says, means the growth in the end or beginning i.e from year 2015 to 2018 and 2021 to 2024 is more for the UK.

Variable 2 (Electric Charging points): Norway has the highest mean, SD and IQR compared to others, which shows very high accelerated growth. The range is seems reasonable for all countries, growth has been exponential. IQR & SD is highest for Norway, but similar for France and Germany, which means the growth has spread through almost equally.

EV Sales varies more than Electric charging points.

3 Distribution Shapes

Plot histograms and compare with a Normal curve for the same variables.

▼ Show code

```
# Histogram and normal curve for ev sales
hist(filtered_ev_data$value/100000,
      probability = TRUE,
      main = "Distribution of ev sales(per 10k)",
      xlab = "ev_sales",
      col = "lightblue",
      ylim=c(0,0.6),
      border = "white")
curve(dnorm(x,
            mean = mean(filtered_ev_data$value/100000, na.rm = TRUE),
            sd = sd(filtered_ev_data$value/100000, na.rm = TRUE)),
      col = "red", lwd = 2, add = TRUE)

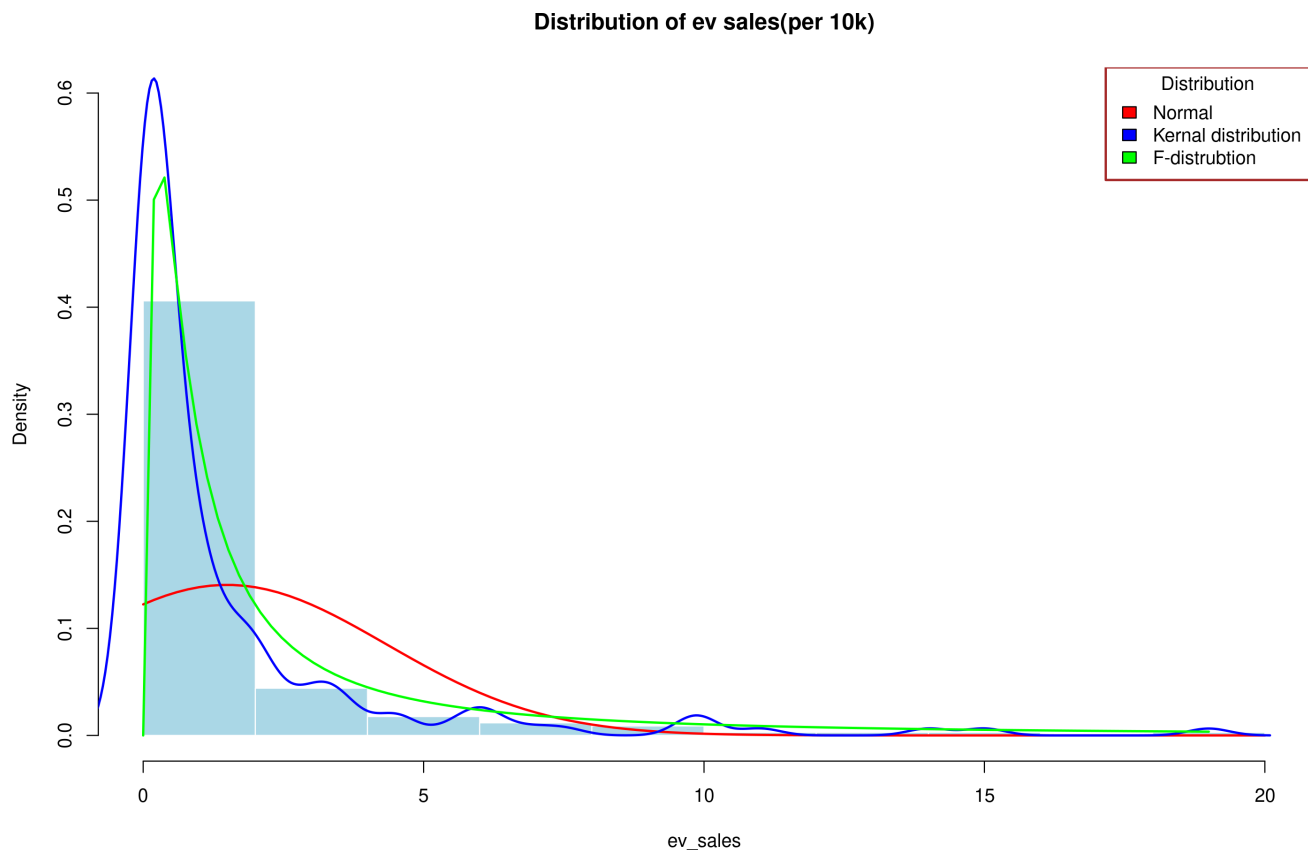
lines(density(filtered_ev_data$value/100000), col = "blue", lwd = 2, add = TRUE)

curve(df(x, df1 = 6, df2 = 1.5), from = 0, to = max(filtered_ev_data$value/100000),
      col = "green", lwd = 2, add = TRUE)
legend(x = "topright", box.col = "brown",
      box.lwd = 2, title="Distribution",
```

```

legend=c("Normal", "Kernal distribution","F-distrubtion"),
fill = c("red","blue","green"))

```



▼ Show code

```

# Histogram and normal curve for electric charging points
ec_wide <- filtered_ec_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)

ec_long <- ec_wide %>%
  pivot_longer(
    cols = -region_country,
    names_to = "year",
    values_to = "value"
  ) %>%
  mutate(year = as.numeric(year))

hist(ec_long$value/10000,
     probability = TRUE,
     main = "Distribution of electric charaging ports(per 10k)",
     xlab = "ec_points",
     col = "lightblue",
     ylim=c(0,1),
     border = "white")

```

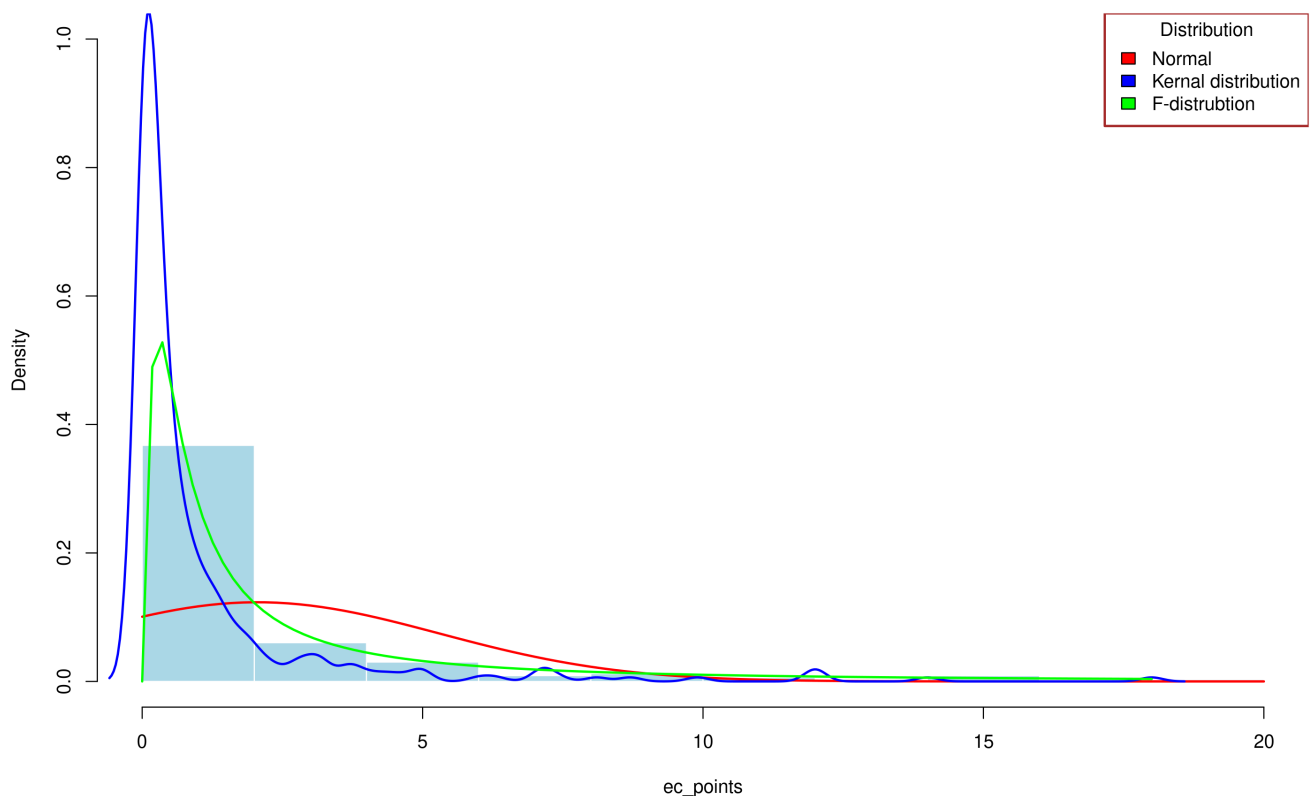
```

curve(dnorm(x,
            mean = mean(ec_long$value/10000, na.rm = TRUE),
            sd = sd(ec_long$value/10000, na.rm = TRUE)),
      col = "red", lwd = 2, add = TRUE)

lines(density(filtered_ec_data$value/10000), col = "blue", lwd = 2, add = TRUE)
curve(df(x, df1 = 6, df2 = 1.5), from = 0, to = max(filtered_ec_data$value/10000),
      col = "green", lwd = 2, add = TRUE)
legend(x = "topright", box.col = "brown",
      box.lwd = 2, title = "Distribution",
      legend = c("Normal", "Kernal distribution", "F-distrubtion"),
      fill = c("red", "blue", "green"))

```

Distribution of electric charging ports(per 10k)



▼ Show code

```

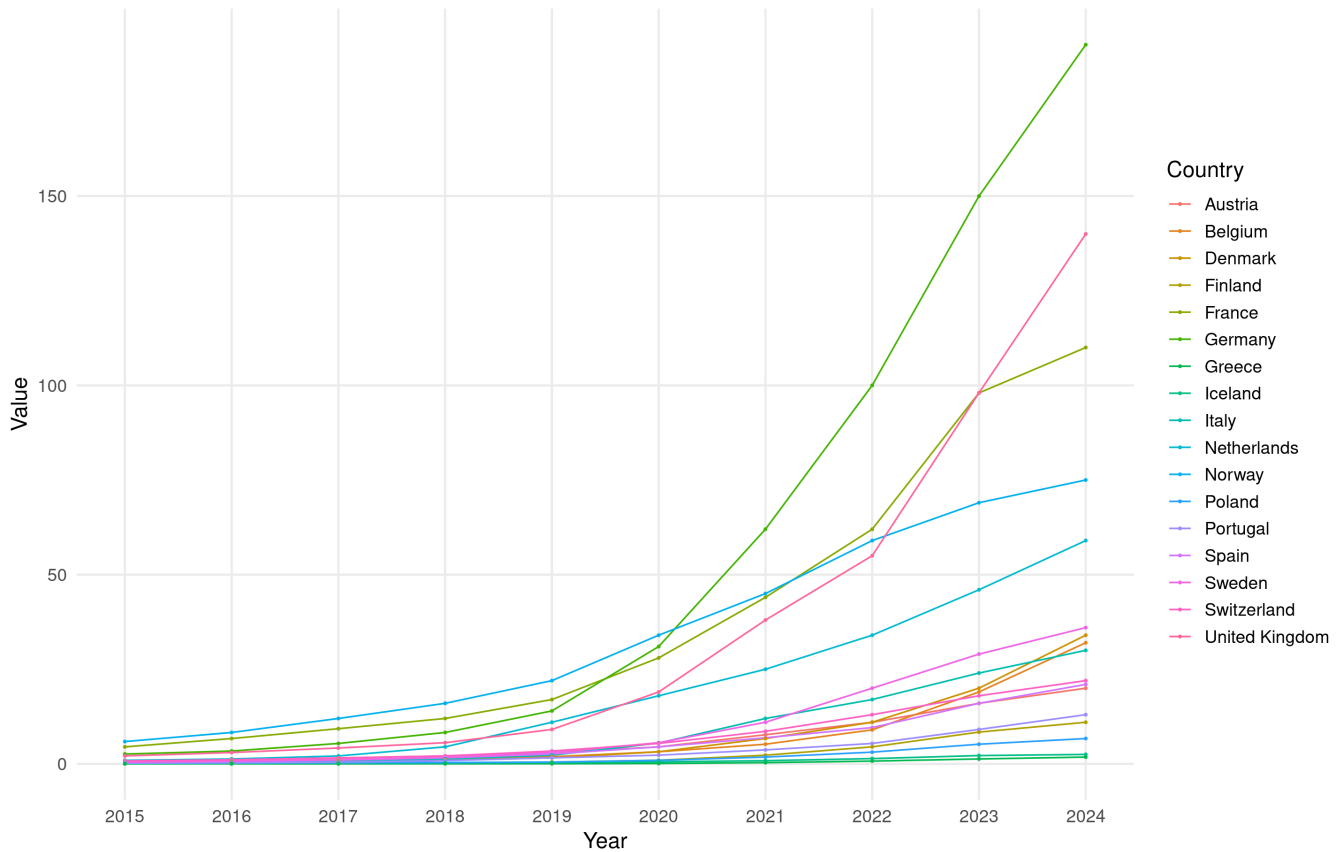
#Line curve for electric sales
ggplot(filtered_ev_data, aes(x = year, y = value/10000, color = region_country, group
  geom_line(size = 0.5) +
  geom_point(size = 0.5) +
  scale_x_continuous(breaks = c(min(filtered_ev_data$year):max(filtered_ev_data$year))
  theme_minimal(base_size = 14) +
  labs(
    title = "EV Metrics Over Time by Country",
    x = "Year",
    y = "Value",
    color = "Country"
  ) +
  theme(
    legend.position = "right",

```

```

plot.title = element_text(face = "bold", size = 16),
panel.grid.minor = element_blank()
)

```

EV Metrics Over Time by Country

▼ Show code

```

ec_wide <- filtered_ec_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_wider(names_from = year, values_from = value)

ec_long <- ec_wide %>%
  pivot_longer(
    cols = -region_country,
    names_to = "year",
    values_to = "value"
  ) %>%
  mutate(year = as.numeric(year))

ggplot(ec_long, aes(x = year, y = value/10000, color = region_country, group = region_
  geom_line(size = 0.5) +
  geom_point(size = 0.5) +
  scale_x_continuous(breaks = c(min(ec_long$year):max(filtered_ec_data$year))) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Electric charging stations over Time by Country",
    x = "Year",

```

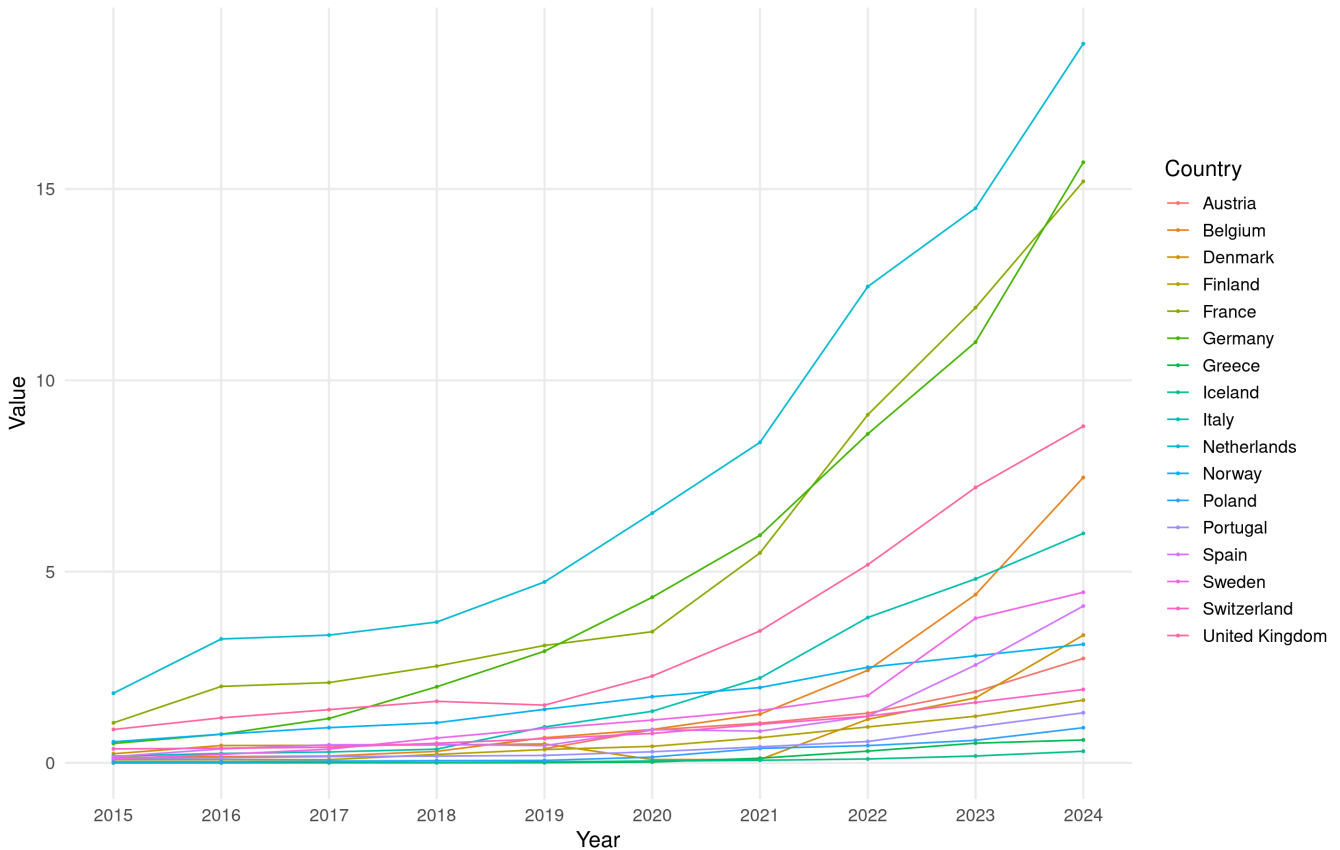


```

y = "Value",
color = "Country"
) +
theme(
  legend.position = "right",
  plot.title = element_text(face = "bold", size = 16),
  panel.grid.minor = element_blank()
)

```

Electric charging stations over Time by Country



Comment (2–3 lines): Is each variable roughly Normal, skewed, or heavy-tailed? Does the red curve fit well? Any gaps or outliers?

Variable 1 (EV Sales): The data is highly right skewed with long tail. It shows the data is accelerated growth in the later years. it is obvious for EV sales , as the adoption to EV in the beginning is lower and grows rapidly over time as seen the line graph. The normal curve does not fit the data very well. It underestimates the density near zero and overestimates it in the tail. There are no visible gaps or outliers.

Variable 2 (Electric Charging Points): This data also is highly right skewed with long tail. It shows the data is accelerated growth in the later years. it is obvious for electric charging points , as the adoption to that in the beginning is lower and grows rapidly over time as seen the line graph. The normal curve does not fit the data very well. It underestimates the density near zero and overestimates it in the tail. There are no visible gaps or outliers.

Exploration: F distribution and Kernal density estimates gives you a better fit. F distributed df1 and df2 values have have been manually adjusted to get the best Fit. Data transformation to log scale or Exponential Regression analysis is required for better modeling frameworks

4 First Model (Simple Linear Regression)

Fit a simple linear model (use appropriate variables and reason your choice).

▼ Show code

```
# create a model usinf lm and provide summary
scatter_dat_France_ec <- ec_long%>%
  filter (region_country == "France")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Germany_ec <- ec_long%>%
  filter (region_country == "Germany")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

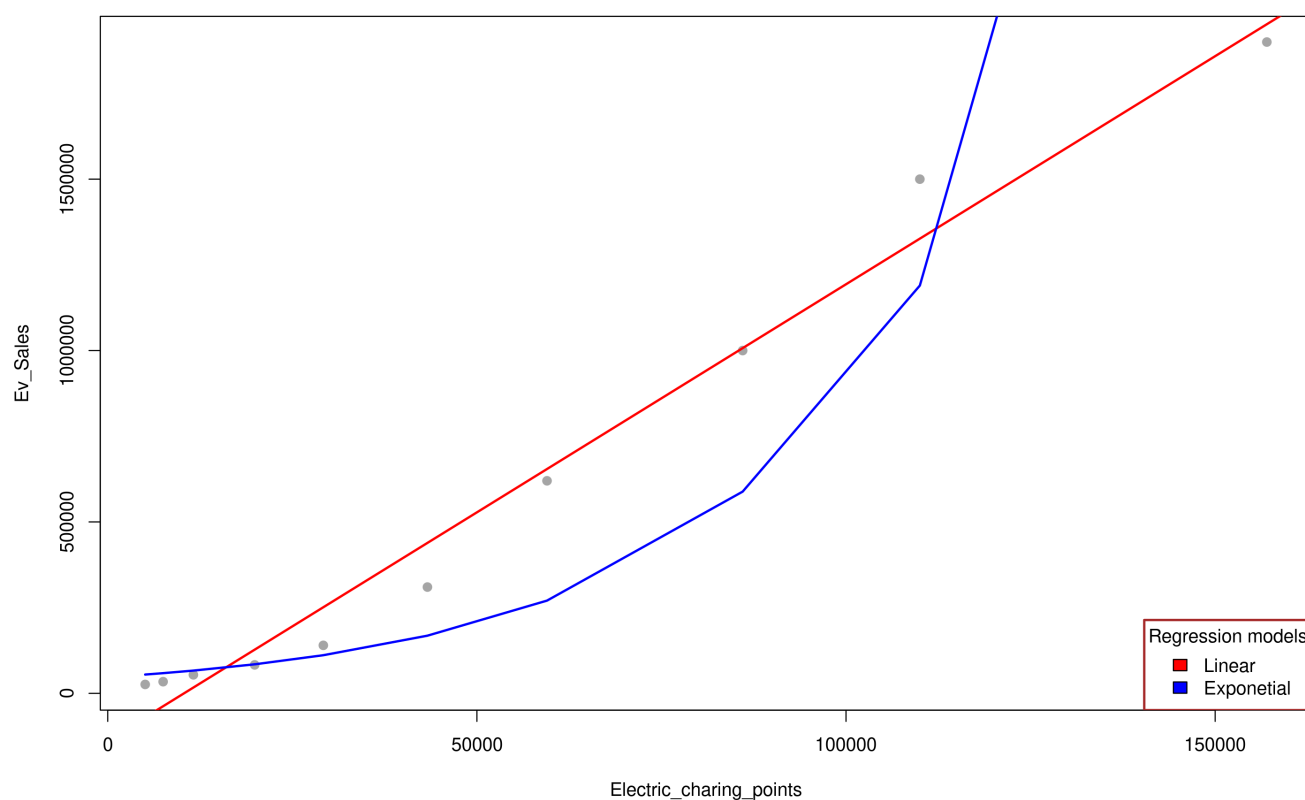
scatter_dat_uk_ec <- ec_long%>%
  filter (region_country == "United Kingdom")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Norway_ec <- ec_long%>%
  filter (region_country == "Norway")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

scatter_dat_Netherlands_ec <- ec_long%>%
  filter (region_country == "Netherlands")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

Cor_ger <- tibble("Electric_charging"= scatter_dat_Germany_ec$value,"EV_sales"= scatter_dat_Germany_ec$value)
plot(Cor_ger$Electric_charging, Cor_ger$EV_sales,
     xlab = "Electric_charing_points",
     ylab = "Ev_Sales",
     main = "Electric_charing_poinEv_Sales vs Ev_Sales in Germany",
     pch = 19, col = "darkgrey")
Ger_model_log <-lm(log(EV_sales) ~ Electric_charging, data = Cor_ger)
Cor_ger$predicted <- exp(predict(Ger_model_log))
Ger_model <- lm(EV_sales~Electric_charging, data = Cor_ger)
abline(lm(EV_sales~Electric_charging, data = Cor_ger), col = "red", lwd = 2)
lines(Cor_ger$Electric_charging, Cor_ger$predicted, col = "blue", lwd = 2)
legend(x = "bottomright", box.col = "brown",
      box.lwd = 2 , title="Regression models",
      legend=c("Linear", "Exponetial"),
      fill = c("red","blue"))
```

Electric_charing_poinEv_Sales vs Ev_Sales in Germany



▼ Show code

```
summary(Ger_model)
```

Call:

```
lm(formula = EV_sales ~ Electric_charging, data = Cor_ger)
```

Residuals:

Min	1Q	Median	3Q	Max
-128847	-50113	-20778	62951	173486

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.374e+05	4.727e+04	-2.907	0.0197 *
Electric_charging	1.331e+01	6.617e-01	20.113	3.9e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 100400 on 8 degrees of freedom

Multiple R-squared: 0.9806, Adjusted R-squared: 0.9782

F-statistic: 404.5 on 1 and 8 DF, p-value: 3.898e-08

▼ Show code

```
summary(Ger_model_log)
```

Call:

```
lm(formula = log(EV_sales) ~ Electric_charging, data = Cor_ger)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.9102	-0.4635	0.1048	0.4556	0.8293

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.076e+01	2.967e-01	36.279	3.65e-10 ***
Electric_charging	2.933e-05	4.153e-06	7.061	0.000106 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

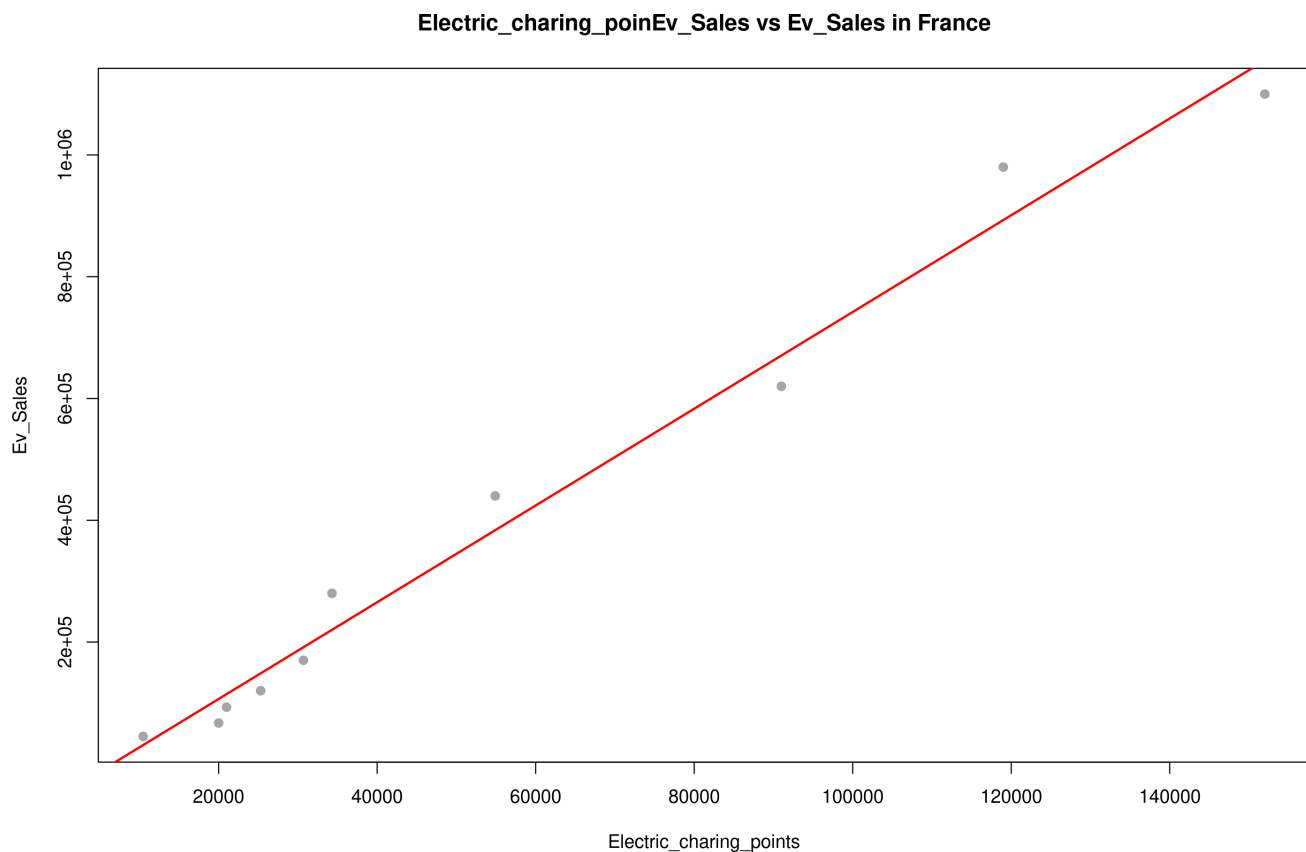
Residual standard error: 0.6303 on 8 degrees of freedom

Multiple R-squared: 0.8617, Adjusted R-squared: 0.8445

F-statistic: 49.86 on 1 and 8 DF, p-value: 0.000106

▼ Show code

```
Cor_fr <- tibble("EV_sales"= scatter_dat_France_ev$value,"Electric_charging"= scatter_
plot(Cor_fr$Electric_charging,Cor_fr$EV_sales,
      xlab = "Electric_charing_points",
      ylab = "Ev_Sales",
      main = "Electric_charing_poinEv_Sales vs Ev_Sales in France",
      pch = 19, col = "darkgrey")
abline(lm(EV_sales~Electric_charging , data = Cor_fr), col = "red", lwd = 2)
```



▼ Show code

```
fr_model <- lm(EV_sales~Electric_charging, data = Cor_fr)
summary(fr_model)
```

Call:

```
lm(formula = EV_sales ~ Electric_charging, data = Cor_fr)
```

Residuals:

Min	1Q	Median	3Q	Max
-55241	-36816	-21516	45667	86934

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.236e+04	2.696e+04	-1.942	0.088 .
Electric_charging	7.945e+00	3.729e-01	21.308	2.47e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 54110 on 8 degrees of freedom

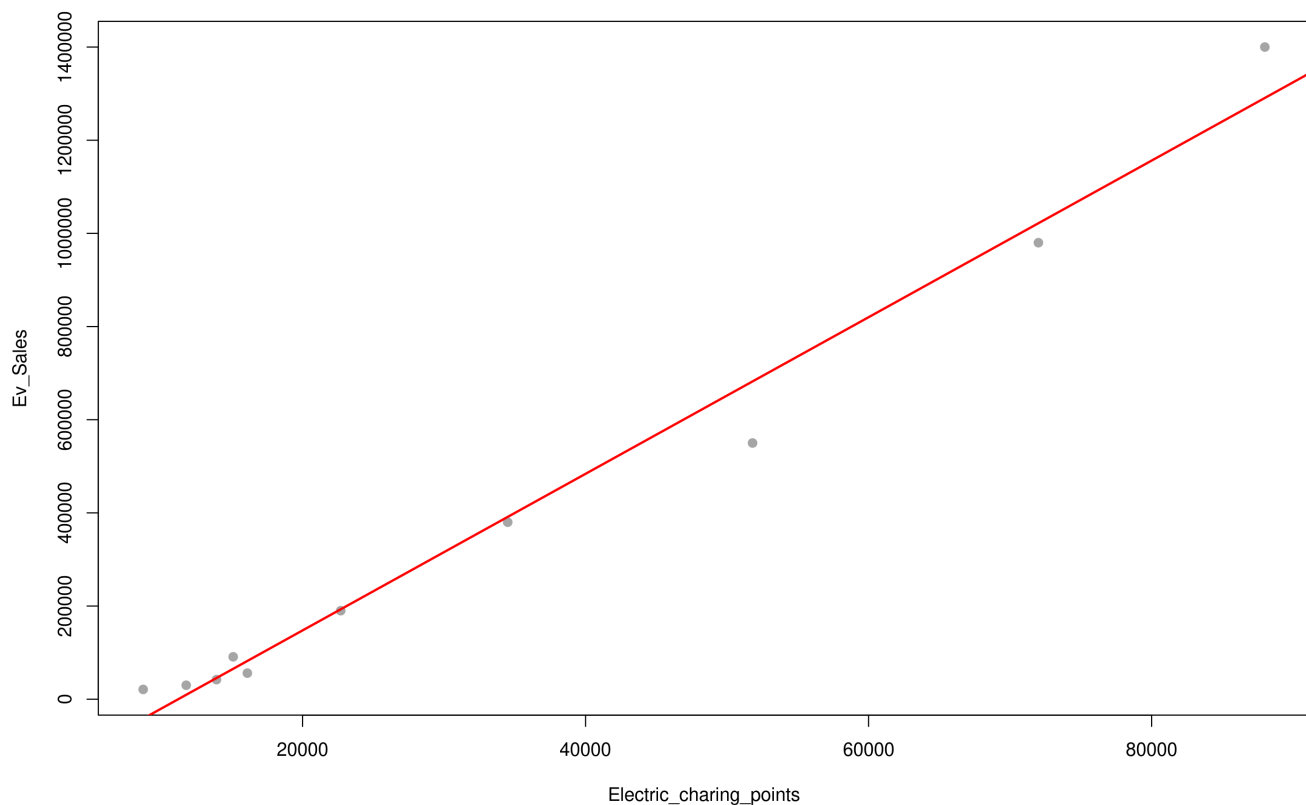
Multiple R-squared: 0.9827, Adjusted R-squared: 0.9805

F-statistic: 454 on 1 and 8 DF, p-value: 2.475e-08

▼ Show code

```
Cor_uk <- tibble("EV_sales"= scatter_dat_uk_ev$value,"Electric_charging"= scatter_dat_
plot(Cor_uk$Electric_charging,Cor_uk$EV_sales,
      xlab = "Electric_charing_points",
      ylab = "Ev_Sales",
      main = "Electric_charing_poinEv_Sales vs Ev_Sales in the UK",
      pch = 19, col = "darkgrey")
abline(lm(EV_sales~Electric_charging , data = Cor_uk), col = "red", lwd = 2)
```

Electric_charing_poinEv_Sales vs Ev_Sales in the UK



▼ Show code

```
uk_model <- lm(EV_sales~Electric_charging , data = Cor_uk)
summary(uk_model)
```

Call:

```
lm(formula = EV_sales ~ Electric_charging, data = Cor_uk)
```

Residuals:

Min	1Q	Median	3Q	Max
-132230	-22436	-3260	24403	109245

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.885e+05	3.465e+04	-5.44	0.000616 ***
Electric_charging	1.681e+01	8.120e-01	20.70	3.11e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 68010 on 8 degrees of freedom

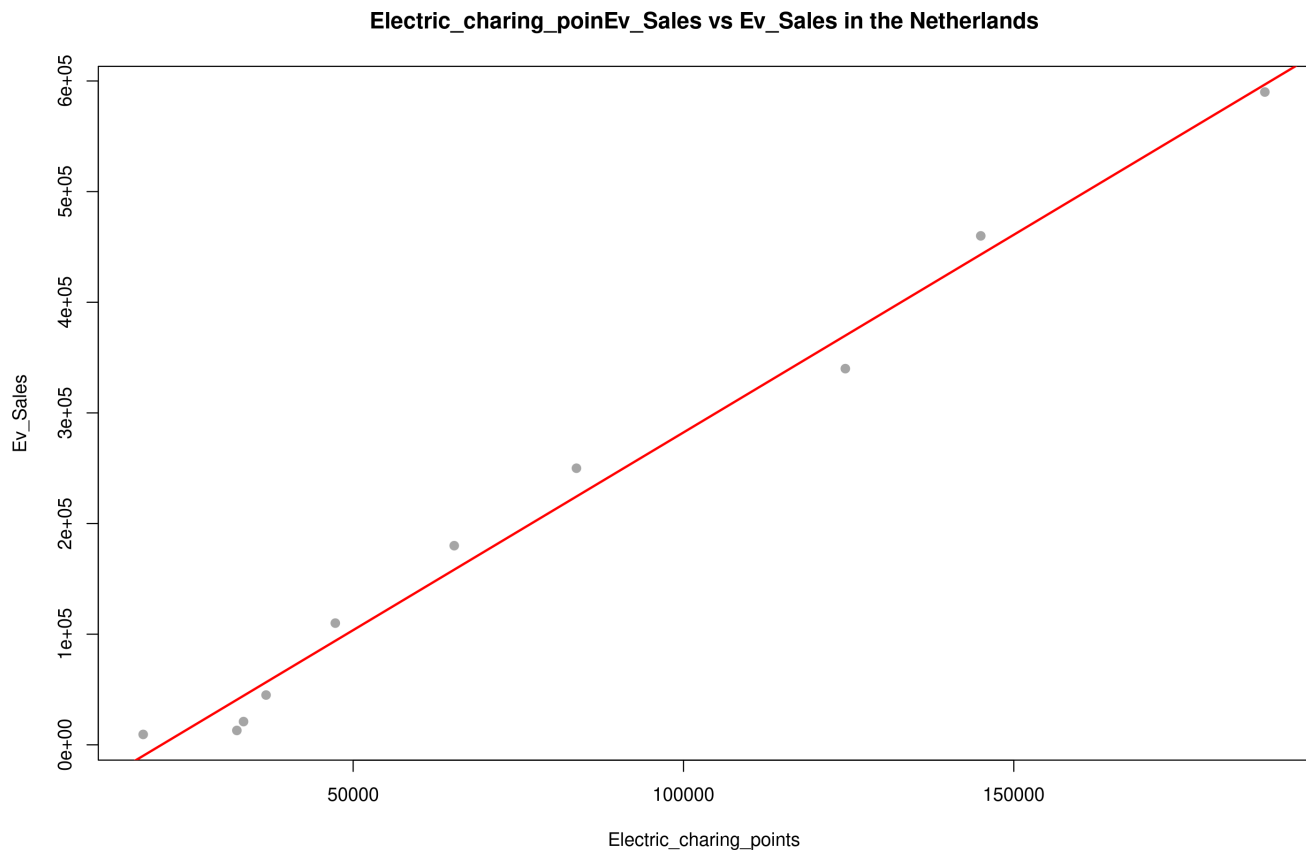
Multiple R-squared: 0.9817, Adjusted R-squared: 0.9794

F-statistic: 428.6 on 1 and 8 DF, p-value: 3.106e-08

▼ Show code

```
Cor_ND <- tibble("EV_sales"= scatter_dat_Netherlands_ev$value,"Electric_charging"= sca
plot(Cor_ND$Electric_charging,Cor_ND$EV_sales,
      xlab = "Electric_charing_points",
```

```
ylab = "Ev_Sales",
main = "Electric_charing_poinEv_Sales vs Ev_Sales in the Netherlands",
pch = 19, col = "darkgrey")
abline(lm(EV_sales~Electric_charging, data = Cor_ND), col = "red", lwd = 2)
```



▼ Show code

```
ND_model <- lm(EV_sales~Electric_charging, data = Cor_ND)
summary(ND_model)
```

Call:

```
lm(formula = EV_sales ~ Electric_charging, data = Cor_ND)
```

Residuals:

Min	1Q	Median	3Q	Max
-29890	-20414	4589	18700	25557

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.503e+04	1.300e+04	-5.772	0.000418 ***
Electric_charging	3.574e+00	1.376e-01	25.974	5.18e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 23520 on 8 degrees of freedom

Multiple R-squared: 0.9883, Adjusted R-squared: 0.9868

F-statistic: 674.6 on 1 and 8 DF, p-value: 5.182e-09

4.1 Model (Exponential Model) - Exploration

Fit a Exponential model to predict the EV sales of germany for the future.

▼ Show code

```
scatter_dat_Germany_ev$t <- scatter_dat_Germany_ev$year - 2015

# Fit exponential model using nonlinear least squares
exp_model <- nls(value ~ a * exp(b * t),
                 data = scatter_dat_Germany_ev,
                 start = list(a = 20000, b = 0.4))

summary(exp_model)
```

Formula: value ~ a * exp(b * t)

Parameters:

	Estimate	Std. Error	t value	Pr(> t)
a	5.051e+04	1.387e+04	3.643	0.00656 **
b	4.098e-01	3.308e-02	12.388	1.68e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 104900 on 8 degrees of freedom

Number of iterations to convergence: 6

Achieved convergence tolerance: 6.546e-06

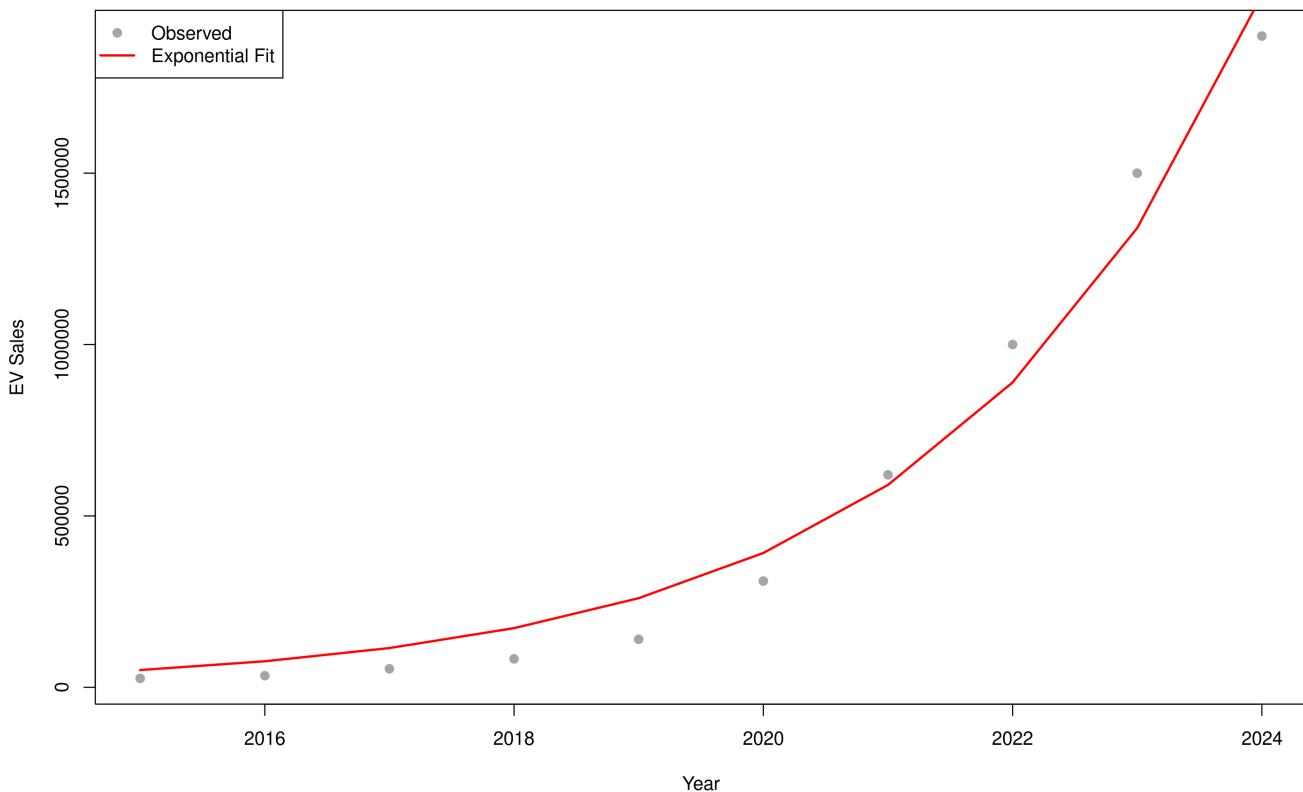
▼ Show code

```
a <- coef(exp_model)[1]
b <- coef(exp_model)[2]

scatter_dat_Germany_ev$Predicted <- predict(exp_model)

# Plot
plot(scatter_dat_Germany_ev$year, scatter_dat_Germany_ev$value, pch = 19, col = "darkgray",
     main = "Exponential Growth of EV Sales in Germany",
     xlab = "Year", ylab = "EV Sales")
lines(scatter_dat_Germany_ev$year, scatter_dat_Germany_ev$Predicted, col = "red", lwd = 2)
legend("topleft", legend = c("Observed", "Exponential Fit"),
     col = c("darkgray", "red"), pch = c(19, NA), lwd = c(NA, 2))
```


Exponential Growth of EV Sales in Germany



Interpretation (3–4 lines): The linear regression model is plotted for EV_sales and Electric charging point for top 5 countries,

For Germany

- Intercept (β_0) = $-1.374e+05$ means for '0' electric charging stations, we have $-1.4e+05$ sales, which is practically impossible. this means our the regression line work for only data points region and cannot be extrapolated to below the data points.
- Slope (β_1) = 13.31 means for every charging station we have 13.31 EV sales. it shows a positive correlation between the 2 variables.
- (Optional) Is the slope statistically significant (p-value)? The p values for the $3.98e-08$ for the slope. it is highly statistically significant.
- (Optional) How much variation does the model explain (R^2)? $R^2 = 0.98$, it implies that changes in charging infrastructure account for nearly all variation in EV sales across years.

All the countries have similar R^2 and p-value, which show that there is a very strong positive linear relationship between charging infrastructure and EV sales. The slopes values are different for every country, and Uk shows highest slope which means the for every charging station we have most EV sales in the Uk. However, the lower region in the data, especially in years 2015 to 2019 , we have very more variation between the model predicted values and the data. So other model types were explored.

Exploration: Exponential model was used, and found the we have better correlation in the lower region in the data, especially in years 2015 to 2019. Overall, we have lesser R^2 value for the model(0.86) but we have better estimate of the intercept(i.e 10). So, if we want to estimate for year less than 2019, exponential model seems to be a better fit.

- Exponential model was used, to correlate between Years and Sales for Germany, and can be used to predict sales for the future.

What is a model?

A **model** is a simplified, mathematical explanation of how things relate or behave.

It is **expressed mathematically** (equation/algorithm), **summarises** relationships, **simplifies** what matters, **generalises** (within limits), and **must be interpretable**.

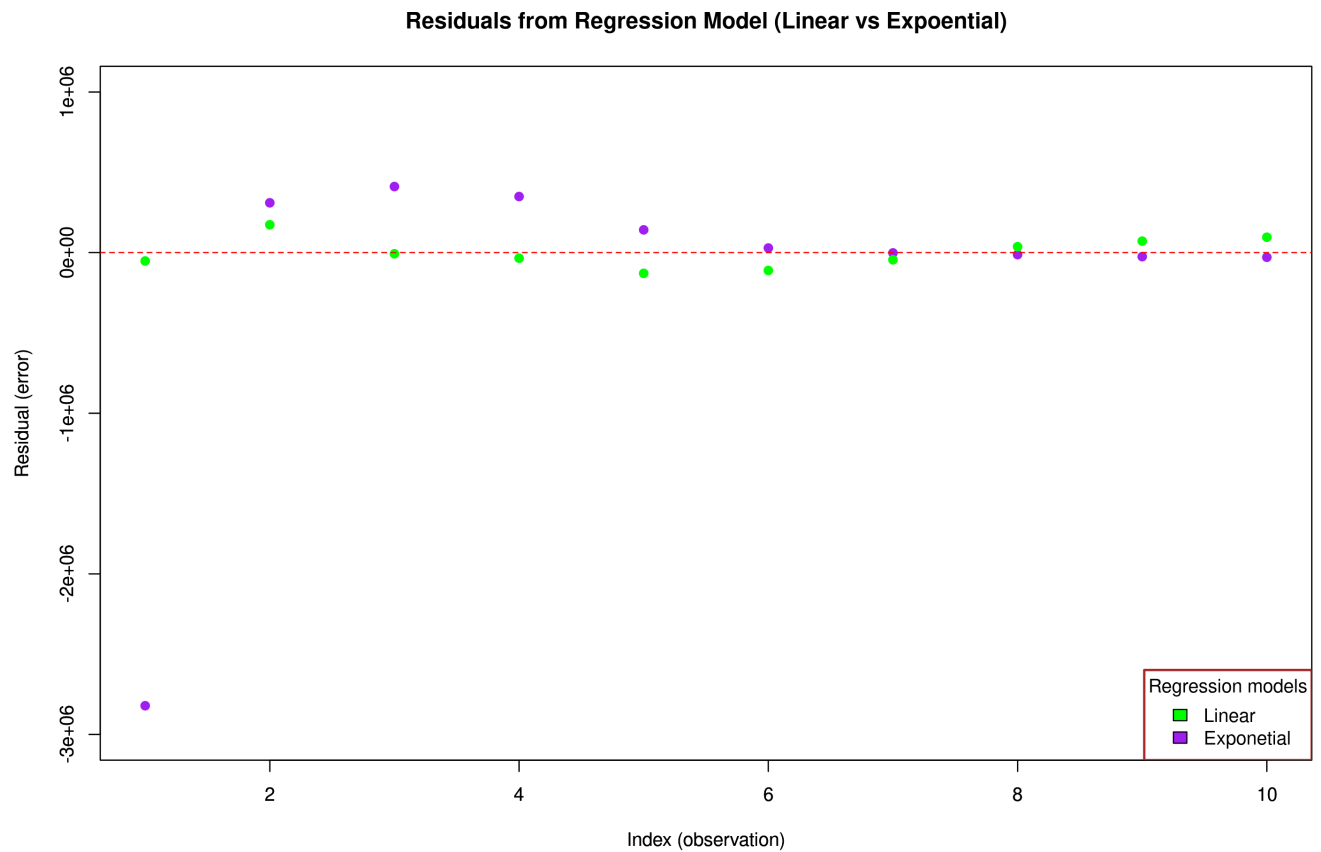
A good model is clear, honest, and useful — not a perfect copy of reality, but a way to think better about it.

5 Model Fit (Residuals)

Residuals are **errors** = observed – predicted. They should look like random noise around 0.

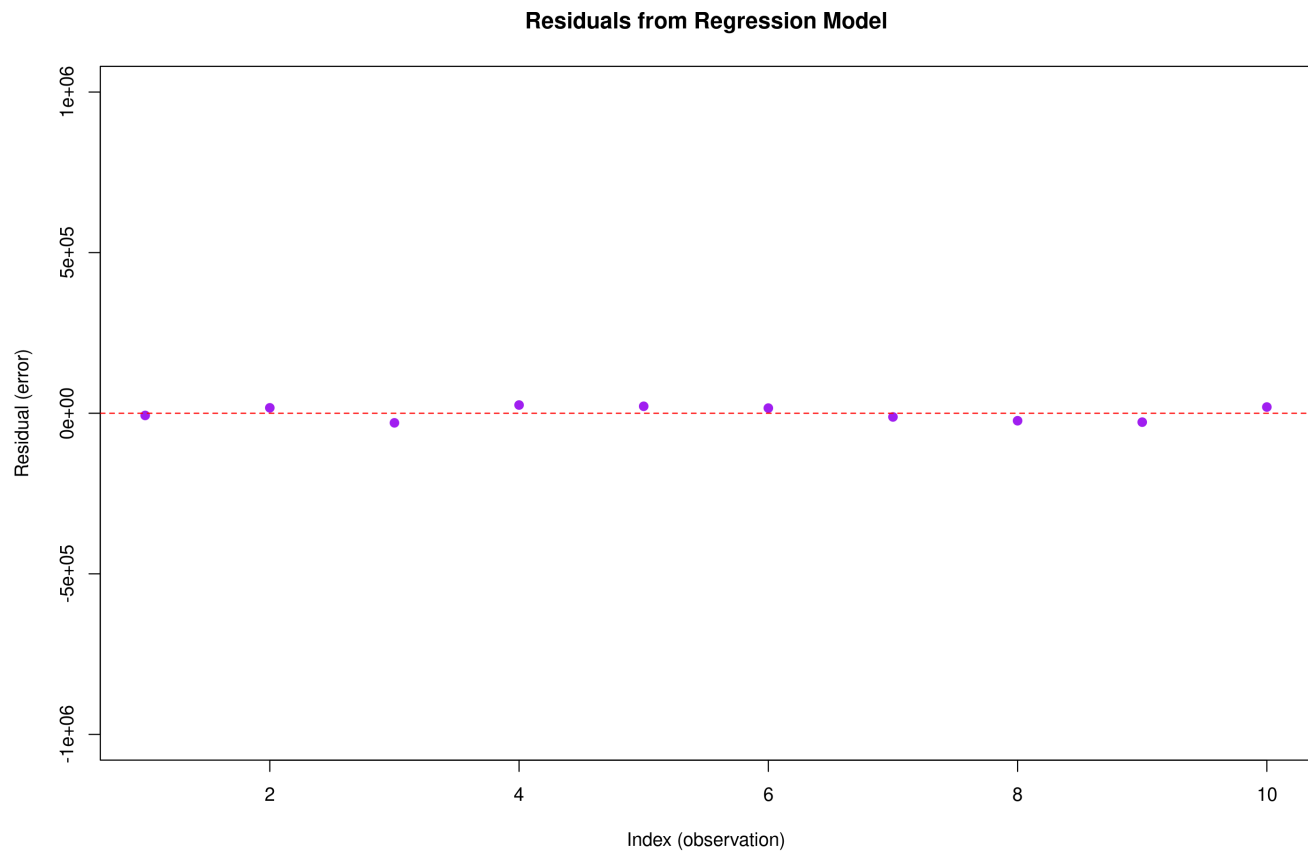
▼ Show code

```
# Plot the residuals including abline
Cor_ger$residuals <- Cor_ger$EV_sales - Cor_ger$predicted
exp_model <- lm(log(EV_sales) ~ Electric_charging, data = Cor_ger)
plot(Cor_ger$residuals,
     main = "Residuals from Regression Model (Linear vs Exponential)",
     ylab = "Residual (error)",
     xlab = "Index (observation)",
     col = "purple", pch = 19,
     ylim=c(-3000000,1000000))
abline(h = 0, col = "red", lty = 2)
par(new=TRUE)
plot(Ger_model$residuals,
     ylab = "Residual (error)",
     xlab = "Index (observation)",
     col = "Green", pch = 19,
     ylim=c(-3000000,1000000))
abline(h = 0, col = "red", lty = 2)
legend(x = "bottomright", box.col = "brown",
      box.lwd = 2, title="Regression models",
      legend=c("Linear", "Exponential"),
      fill = c("green","purple"))
```



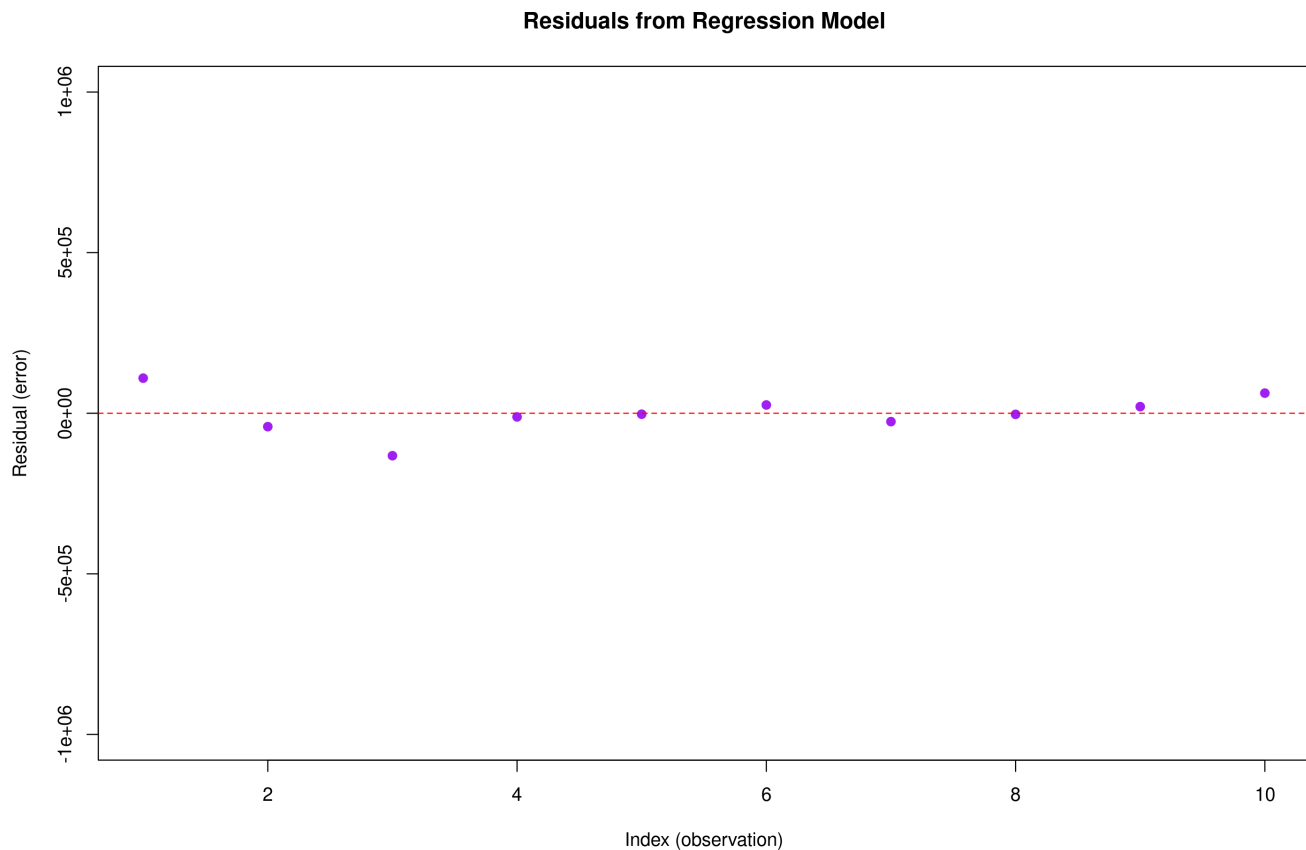
▼ Show code

```
plot(ND_model$residuals,  
     main = "Residuals from Regression Model",  
     ylab = "Residual (error)",  
     xlab = "Index (observation)",  
     col = "purple", pch = 19,  
     ylim=c(-1000000,1000000))  
abline(h = 0, col = "red", lty = 2)
```



▼ Show code

```
plot(uk_model$residuals,  
     main = "Residuals from Regression Model",  
     ylab = "Residual (error)",  
     xlab = "Index (observation)",  
     col = "purple", pch = 19,  
     ylim=c(-1000000,1000000))  
  
abline(h = 0, col = "red", lty = 2)
```



Comment (2–3 lines): Are residuals roughly centred around 0 and randomly scattered? Any patterns or extreme points? What does this suggest about model fit?

The residuals graphs show the for all countries, the residuals are mostly centered around 0. They are not randomly scattered but follow a curve or a systemic pattern (low and mid points negative and extremes have positive residuals) which show a non linearity and can be better fitted with a quadratic or exponential model. There are no extreme values which suggests no outliers.

Exploration: Exponential model which was used in the previous section, the residuals were compared with the linear model only in the case of Germany. It clearly highlights our findings that in the initial years, i.e from 2015 to 2019, the residuals from the exponential model are better than linear model. But for later years, we have bigger difference from base for the exponential model than linear model. This proves that, to estimate for years less than 2019, exponential model is a better fit.

6 Group Reflection

Keep answers short and specific (1–2 sentences each).

Question	Reflection
Distributions: What did shapes reveal about variation or consistency?	Both variables are highly right skewed with long tail which shows accelerated growth in the later years. The normal curve does not fit the data very well, F distributed with adjusted df1 and df2 values have the best Fit.

Question	Reflection
Model: What does your regression line suggest about the relationship?	All the countries in top 5 have good R^2 (0.98) and p-value ($p < 0.001$), which show that there is a very strong positive linear relationship between charging infrastructure and EV sales. Uk has the best slope which suggest better adoption of EV due to charging infrastructure. The residuals shows a systemic pattern which show a non linearity and can be better fitted with a quadratic or exponential model. From residual analysis it was found, to estimate for years less than 2019, exponential model is a better fit.
Learning: What was the main thing you learned about your dataset this week?	- Data Spread: IQR and SD how it is significant, some countries had similar IQR but extremely different SD it suggested growth in the middle period was almost same but at extremes had different growth rate. - Distributions: Data sets with accelerated growth over years, have genrally highly right skewed data. Normal curve dont fit well for these skewed types. F-distribution seems a better fit and how df1 and df2 can be adjusted to fit our data. - Model: p-value shows statistical significance between two variables. R^2 tells how well your is model behaves. Slopes for different datasets be can be predictor to show which is better. Residual Plots can tell us about the non linearity and systemic bias in the mode Different models can be is used to analyse different regions of the data, depending on the residual plots.

7 Expected Outputs

Your file should include (at least): - Two distribution plots - One scatterplot with fitted regression line - One residual plot - Regression summary with brief interpretation - Completed reflection table

8 You are free to explore further along modeling (e.g., trying different variable combinations, or trying multiple regression, or different models available in R libraries).

9 You may also explore along p-values and R^2 .

Week 5 – Bias, Fairness, Ethics & Reproducibility

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PUBLISHED
November 17, 2025

Overview

What this file is for

Identify common sources of bias across the data-science pipeline.

Explain why fairness requires balancing trade-offs rather than removing bias completely.

Discuss core ethical principles using the FATE framework — Fairness, Accountability, Transparency, Explainability — and connect them to your project context.

Apply practical steps for reproducibility, such as setting random seeds, logging environments.

Reflect on how fairness, ethics, and reproducibility together create trustworthy results and audit-ready evidence for your group project.

1 Bias

Stage of bias summary

Stage of workflow	Amplify or reduce	Control Strategies	Verdict
Sampling	Data is mainly mature European countries, omitting emerging markets or incomplete reporting. this may inflate or deflate the correlation between infrastructure and EV growth, since included nations have more mature ecosystems.	Document geographical scope clearly and avoid over generalization to global contexts.	⚠️
Measurements	The variables such as 'EV sales" and "Charging point" are reported by different agencies to IEA, possibly using inconsistent definitions or estimation methods.	Lack of standardization may skew country comparisons.	⚠️
Temporal Bias	There maybe other factors like rapid policy shifts in this period (2015-2024).Lack of	Already had plan to have policy indicators as one of the variable	✅

Stage of workflow	Amplify or reduce	Control Strategies	Verdict
	standardization may skew country comparisons and Ignoring these contextual changes could mis-attributed policy-driven changes to market factors.		
Modeling Bias	Linear regression is not a good fit and can underestimate early stages or overstate long run predictability.	Already tested with exponential growth models; check residuals for structure.	✓
Interpretation Bias	Causation might be bidirectional.	Other statistical studies to shows if it both influencing each other.	✓

2 Exploring Fairness & Trade-offs

Write a few lines in describing how fairness could be interpreted in your project's context, and which kind of fairness —**representation or outcome fairness**— feels most relevant to your goal.

For our project, Fairness primary means how countries and regions are represented in the data set and how our model can be accurate across different countries. The dataset contains data from 17 European nations, but representation is uneven, Large countries such as Germany, France, and the UK have more complete and detailed data, while smaller or emerging EV markets (e.g. Eastern European countries) are under represented or missing entirely. This reflects a representation fairness issue, where some regions have greater visibility and influence in the analysis.

When the data are used predicatively, modelling EV sales as a function of charging infrastructure, fairness also extends to outcome fairness, since the model may perform better for countries with dense, high-quality data and worse for those with limited or inconsistent records.

For this project, representation fairness is the more relevant concern, because the goal is to understand patterns of EV adoption across countries rather than predict outcomes for specific groups. Ensuring fairness thus means clearly acknowledging gaps in regional coverage and avoiding overgeneralizing findings beyond the dataset's represented contexts.

2.1 Checking Groupwise Outcomes

▼ Show code

```
library(tidyverse)
library(readxl)
library(dplyr)
options(dplyr.summarise.inform = FALSE)
knitr::opts_chunk$set(fig.width = 12,
                      fig.height = 8,
                      include = TRUE,
                      echo = TRUE)

# Load your cleaned dataset
filtered_ev_data <- read_csv("data/processed/EV_Europe_BEV_Stock_2015_2024.csv")
filtered_ec_data <- read_csv("data/processed/EC_Europe_BEV_Stock_2015_2024.csv")

# Quick structure check
glimpse(filtered_ev_data)
```

Rows: 170

Columns: 3

```
$ region_country <chr> "Germany", "Germany", "United Kingdom", "France", "Germ...
$ year          <dbl> 2024, 2023, 2024, 2024, 2022, 2023, 2023, 2024, 2023, 2...
$ value         <dbl> 1900000, 1500000, 1400000, 1100000, 1000000, 980000, 98...
```

▼ Show code

```
glimpse(filtered_ec_data)
```

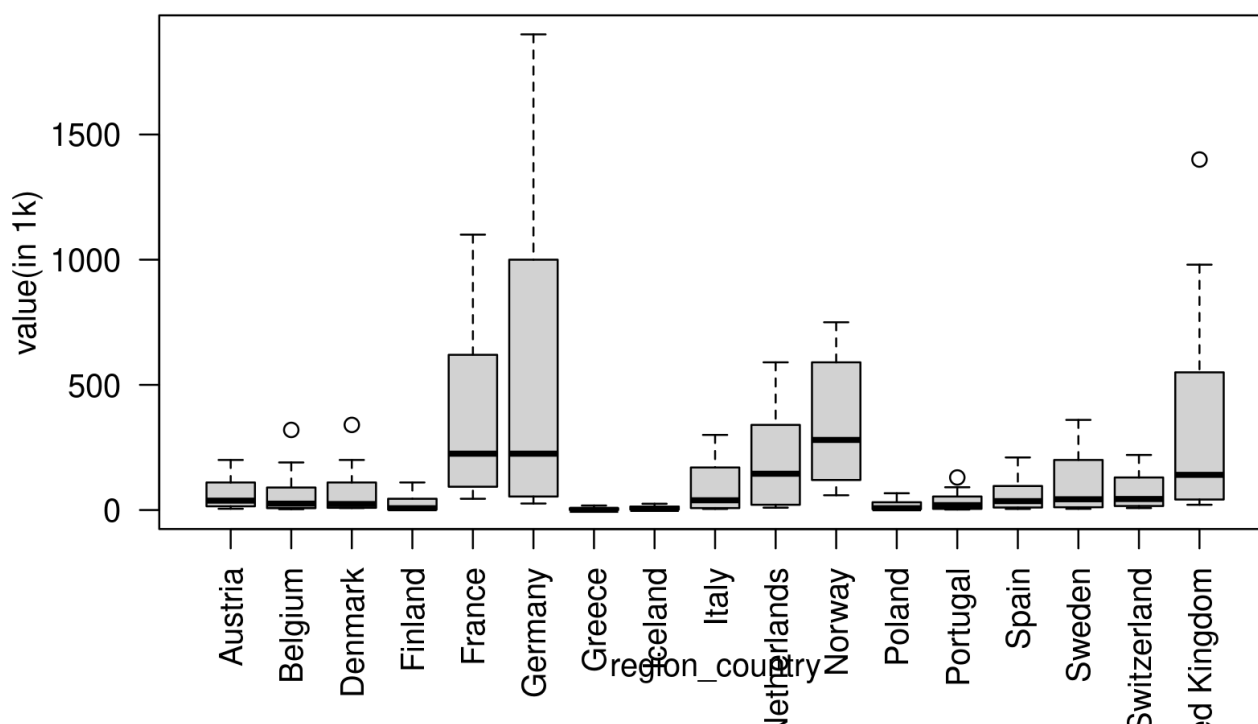
Rows: 325

Columns: 3

```
$ region_country <chr> "Netherlands", "Netherlands", "France", "Germany", "Net...
$ year          <dbl> 2024, 2023, 2024, 2024, 2022, 2023, 2023, 2021, 2022, 2...
$ value         <dbl> 180000, 140000, 120000, 120000, 120000, 99000, 87000, 8...
```

► Show code

EV Stock in different by Region



▼ Show code

```
scatter_dat_Greece_ev <- filtered_ev_data%>%
  filter (region_country == "Greece")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))
scatter_dat_poland_ev <- filtered_ev_data%>%
  filter (region_country == "Poland")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))
scatter_dat_Iceland_ev <- filtered_ev_data%>%
  filter (region_country == "Iceland")%>%
  select(region_country, year, value) %>%
  arrange(desc(year))

summary_BTm3_ev <- tibble(
  Country = c("Iceland","Poland","Greece"),
  Max = c(max(scatter_dat_Iceland_ev$value),max(scatter_dat_poland_ev$value),max(scatter_dat_Greece_ev$value)),
  Min = c(min(scatter_dat_Iceland_ev$value),min(scatter_dat_poland_ev$value),min(scatter_dat_Greece_ev$value)),
  Mean = c(mean(scatter_dat_Iceland_ev$value),mean(scatter_dat_poland_ev$value),mean(scatter_dat_Greece_ev$value)),
  Median = c(median(scatter_dat_Iceland_ev$value),median(scatter_dat_poland_ev$value),median(scatter_dat_Greece_ev$value)),
  Std_Dev = c(sd(scatter_dat_Iceland_ev$value),sd(scatter_dat_poland_ev$value),sd(scatter_dat_Greece_ev$value)),
  IQR = c(IQR(scatter_dat_Iceland_ev$value),IQR(scatter_dat_poland_ev$value),IQR(scatter_dat_Greece_ev$value))

summary_BTm3_ev
```

Country	Max	Min	Mean	Median	Std_Dev	IQR
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Iceland	25000	570	8403.0	4450	8959.492	10750.0

Country	Max	Min	Mean	Median	Std_Dev	IQR
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Poland	67000	2500	19410.0	7100	23233.666	24500.0
Greece	18000	97	4397.7	785	6384.241	6162.5

3 rows

How could unequal proportions or outcomes for that variable create fairness concerns later? In our dataset, the variable “Country” divides the data into subgroups such as Germany, France, the UK, Norway, and the Netherlands. This variable is central to the analysis because EV sales, charging infrastructure, and policy incentives differ significantly between these countries.

However, the distribution across countries is uneven as seen in the box plot, Germany and France have far more complete and higher-magnitude data points compared to smaller markets like the iceland or Greece. This creates a representation imbalance, where countries with more available or larger-scale data disproportionately influence the model outcomes.

▼ Show code

```
# compare average EV sales between two random groups of countires in europe
set.seed(7) # Ensures everyone creates the same split
data_cat <- filtered_ev_data|> mutate(group = sample(c("A", "B"), n(), replace = TRUE))

# Compare average ev_sales in each group
data_cat |>
  group_by(group) |>
  summarise(avg_EV_sales = mean(value, na.rm = TRUE))
```

group	avg_EV_sales
<chr>	<dbl>
A	142452.4
B	154709.6

2 rows

▼ Show code

```
# compare average EV sales between two random groups of countires in europe
set.seed(21) # Ensures everyone creates the same split
data_cat1 <- filtered_ev_data|> mutate(group = sample(c("A", "B"), n(), replace = TRUE))

# Compare average ev_sales in each group
data_cat1 |>
  group_by(group) |>
  summarise(avg_EV_sales = mean(value, na.rm = TRUE))
```

group	avg_EV_sales
<chr>	<dbl>
A	171220.4
B	129156.7

2 rows

Describe specifically how you might detect or document this imbalance. As seen in the seed test, the seed test is done for the EV sales for two different group of countries. The values varies a lot. in the first group 1 B is larger than B anf in the second group A is larger than B.

If we later use the model to make predictions or comparisons, it might overestimate or underestimate trends for underrepresented countries, reducing fairness in the interpretation of results. For example, a model trained mainly on Germany's data might assume the same growth trajectory for Norway, despite very different policy contexts and market sizes. But because the goal is to understand patterns of EV adoption across countries rather than predict outcomes for specific groups. Thus means clearly acknowledging gaps in regional coverage and avoiding overgeneralizing findings beyond the dataset's represented contexts.

We can document, visualize, and clarify these disparities, using statsitics and box plots as shown above, though we cannot fully correct the data imbalance itself.

3 Human Bias in the Loop

3.1 Confirmation Bias

Try writing at least one possible hypothesis your group could test in your dataset “— for instance, “students with higher study hours tend to have higher grades.”

We have already created the required hypothesis for our project in week 2 analysis and we are gonna copy it in here.

The key objective of this study is try to quantify the influence of demographic, economic, and infrastructural variables on EV adoption rates. The main hypothesis of our study which was established through extensive literature surveys where EV adoption rate determinants was investigated. Smith et al. (2022) found that income and urban density are significant predictors, while charging infrastructure availability strongly influences consumer decisions.

H1 Higher median income positively correlates with EV adoption.

H2 Greater density of public charging stations leads to higher adoption.

H3 Regions with higher fuel prices have faster EV uptake.

3.2 Automation Bias

What could make its predictions unreliable or biased? How might a human sense-check those predictions before acting on them?

In our project, we use a linear regression model to predict or explore relationships between EV sales and electric charging infrastructure. While this model helps identify correlations and trends, its predictions could become unreliable or biased for several reasons:

Data imbalance: As explained in the previous sections, some countries like Germany & France has more data and higher sales values than others, which can skew the regression line toward dominant patterns that don't generalize to smaller markets.

Measurement or reporting errors: Also explained in above sections, Inconsistent data collection between countries.

Non-linearity: The relationship is not linear and this might cause over or under reporting.

We as analyst can do systemic checks like,

- For Model correctness: Plot residuals and diagnostic graphs to check if the model systematically over- or under-predicts for specific countries.
- Compare predicted values with real-world trends and previous years' data to spot implausible spikes or drops.
- Cross-check with alternative models (like exponential model) to test robustness.

3.3 Publication Bias

Are there any results or analyses your group might feel less inclined to share because they look weak, messy, or “off-pattern”? How could documenting or briefly mentioning them actually increase trust in your conclusions?

- We have decided not to do EV_Sales Vs Charging point with all countries together as each countries might behave differently. we have conducted the analysis only for the top 5 most EV-sales and we found “charging point” is strongly correlated to “EV sales” and made conclusions on the charging point. it has to be extended to all countries , although we have very less vales for the bottom few countries.
- Linear Regression model has to be extended to all countries , although, Lower countries might have messy R2 values and has to be reported.
- We need to extend the analysis to other parameters likes income levels, policy incentives, fuel prices, and population density. Excluding these factors can lead to misleading coefficients. And also check for interdependence.

We would mention this in our final report, demonstrating that the group has evaluated limitations, considered alternative explanations, and avoided overstating causal links.

4 The FATE Framework

Principle	Core idea	Example in practice
Fairness	Avoid systemic disadvantage or unfair treatment of any group.	In our project, we will demonstrate Fairness by examining EV adoption patterns across all countries in europe and years, ensuring that no single region dominates the analysis. We will also check for imbalances in data availability that might bias our interpretation of global

Principle	Core idea	Example in practice
		EV trends and document them in our report.
Accountability	Be answerable for the outcomes of your work.	In our project, we will demonstrate accountability by attributing data sources to the IEA and noting any uncertainties or assumptions about data coverage. This helps ensure that users of our outputs understand the limits of our interpretation.
Transparency	Make your process visible and understandable.	In our project, we have demonstrated transparency by clearly documenting all data-cleaning, transformation, and modelling steps in our README file.
Explainability	Ensure your methods can be explained in plain language.	We will demonstrate Reproducibility by sharing our full R code , output files and README file, enabling others to replicate our data processing and regression results.

5 Ethical Domains in Practice

Which of these domains — privacy, transparency, representation, environment, or society — connects most directly to your project? Note any concrete action(s) your team could take this semester to reduce risk or increase fairness in the context.

The domain that most directly connects to our project is representation. Our analysis depends on EV adoption data that varies significantly between regions and years, which may reflect unequal reporting or data collection capacity. Some countries or emerging markets are underrepresented compared to regions in Europe, potentially biasing our conclusions about global trend analysis.

To reduce this risk and improve fairness, we can:

- Acknowledge data gaps in underrepresented regions (e.g., missing data for low-income countries).
- Analyze each country data separately to account for population or vehicle market size, reducing bias from unequal dataset sizes.
- Document data provenance and any filtering steps transparently in our report and README.

6 Reporducibility

Establish reproducibility details to a text file.

▼ Show code

```
# (3) Save reproducibility details to a text file
sink("session-info.txt") # redirect console output to a file
cat("=== Reproducibility Log ===\n")
cat("Random seed used: 21 & 7\n")
cat("Date:", Sys.Date(), "\n\n")
sessionInfo()           # records R version, OS, and packages
sink()                  # stop redirection

# Check that the log file was created
file.exists("session-info.txt")
```

[1] TRUE

You’ve just learned about reproducibility log that captures your software environment, seed, and key model steps. Take time to record your thoughts in week5_BER.qmd

1. Implementation:

- We dont have to seed in our original analysis. But for the sake of this notebook, we have had seeds for analysis and it was note in the session log.
- We have session log for this week file named “session-info.txt”.
- we have documents the packages using sessioninfo() for this week.
- updated README with this weeks analysis.

2. Next Step:

- version control: Clearly record the version of the EV dataset you are using and Qmd analysis version for future analysis.
- Data changes log: Note any preprocessing steps in a changelog or readme file.
- Clear commit messages: Include details about what changed.
 - Seed random operations: If you are splitting data or sampling for model training, set random seeds in code so results can be reproduced.
- Data validation checks: Implement simple checks to catch inconsistencies early.

3. Audit Connection:

- Dataset snapshots: Stored EV datasets with clear version numbers (like EV_data_v2.1.csv) and download dates.

- **Change log:** A log describing updates, corrections, and preprocessing applied to the raw dataset.
Clean, well-documented scripts or notebooks: Every preprocessing step, feature engineering, and analysis method is clearly explained.
- **Script naming conventions:** Scripts reflect their purpose (e.g., preprocess_battery_data.qmd, analyze_charging_patterns.qmd).
Random seeds included: Any sampling or splitting uses fixed seeds, ensuring reproducible results.
- **Commit history:** Git commits with clear, descriptive messages that reflect the changes made to code, data, or analyses.
- **Intermediate results:** Already we have our saved outputs (e.g., cleaned CSVs, Analysis tables) in separate folder.
- **Figures and tables linked to code:** Each plot or table references the script and dataset version that generated it.
- **Step-by-step README or audit guide:** Explains how someone can rerun your analysis from raw data to final results.

4. Long-Term Habit:

- All the points in step 3 need to be implemented going forward.
- Dataset snapshots
- Change log
- Script naming conventions
- Commit history
- Intermediate results
- Step-by-step README

7 Group Reflection

Question	Reflection
Bias:	Our data set reveal that we have a risk of representation bias for the different countries i.e. representation is uneven, Large countries such as Germany, France, and the UK have more complete and detailed data, while smaller or emerging EV markets (e.g. Eastern European countries) are under represented or missing entirely. we cannot solve this as it part of the structure of the data, but we have documented in our qmd and will be also documenting in our final report.
Fairness:	As explained above, we have a risk of representation bias and we need to be careful in our analysis. we plan to do each country separately and account to say that we have strong correlation in particular country rather than generalize it for all european countries.

Question	Reflection
Ethics	From the ethical standpoint, we have a risk of representation again. Acknowledge data gaps in underrepresented regions. Analyze each country data separately to account for population or vehicle market size, reducing bias from unequal data set sizes. Document data provenance and any filtering steps in our README file.
Reproducibility	For trace-ability, we have added session logs, updated README files, added Version control for data set, use clear commit messages. Other things mentioned will be implemented from next week qmd.

Readme_proj

AUTHOR

Group ID: Ods-team-50

PUBLISHED

November 18, 2025

1 Dataset: Global electric vehicle (EV) trends(Cleaned)

Source: <https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer>

Last updated: 31 Jul 2025

License : CC BY 4.0

Dataset title: EVDataexplorer2025

Creator / Publisher: IEA Global

Version / Release date: 31st July'25 Version 1.2

Access URL or API endpoint: <https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer>.

Date retrieved (YYYY-MM-DD): 2024-10-15

Retrieval method (script + function call / manual download): Manual Download

Expected update cadence: 31st July'26

License (name + URL): CC by 4.0 License; <https://www.iea.org/terms/creative-commons-cc-licenses>

Required attribution text (verbatim): The IEA's preferred attribution is: IEA 202[X]; [Title name], [URL], License: [relevant CC license e.g. CC BY 4.0]

Known caveats (units, derived fields, coverage gaps): Schema Evolution: There are periodic updates with new variable names, structure can cause schema inconsistency across years.

Schema snapshot location (e.g., data/external/_schema_preview.csv): data/raw/EVDataExplorer2025.xlsx

Description: - The data frame contains 9 columns

- The data frame has 16,436 row (observations)

The data frame contains 9 columns, which includes 'region_country', 'category', 'parameter', 'mode', 'powertrain', 'year', 'unit', 'value', 'Aggregate group'. The main parameters we are interested are the region_country, year and unit.

Cleaning Step:

The data has to be cleaned and transformed for the next step of analysis. We need to extract just the EV sales, Electric charging points, FCEV, PHEV data from 2015 to 2024 for only the European countries. The following stepped were followed,

- Identify the list of European countries and store it in a vector called europe_countries.
- Using filtering process filter out the European countries, and str_detect function filter put only historical, BEV, EV stock , Cars data from year 2015 to 2024.
- Selecting only the required columns
- Cleaning the names of th data from any any white spaces
- Grouping and summary to see the data
- using Pivot wide function to create a table with different years nd countries.

This process has been repeated from different parameter that we plan to study like Electric charging points, FCEV, PHEV.

Analysis : Week 4 Distribution : Normal, F and kernel densities.

Both variables are highly right skewed with long tail which shows accelerated growth in the later years.

The normal curve does not fit the data very well, F distributed with adjusted df1 and df2 values have the best Fit.

Model : Linear and Exponential regression.

All the countries in top 5 have good R^2 (0.98) and p-value ($p < 0.001$), which show that there is a very strong positive linear relationship between charging infrastructure and EV sales.

Uk has the best slope which suggest better adoption of EV due to charging infrastructure.

The residuals shows a systemic pattern which show a non linearity and can be better fitted with a quadratic or exponential model.

From residual analysis it was found, to estimate for years less than 2019, exponential model is a better fit.

Residual plots.

Week 5 : Ethics Bias and Reproducibility - Explored ethics , bias for our data.

- Added Session information.

Files: "EV_Europe_BEV_Stock_2015_2024.csv" "EC_Europe_BEV_Stock_2015_2024.csv"

"EV_Europe_FEV_Stock_2015_2024.csv" "EV_Europe_PHEV_Stock_2015_2024.csv"